POWER STATION - SCALA

13:30 - 14:30

KEEPING THE FABRIC FACTORY OPERATIONAL:
TURNING COST KNOBS AND BUILDING BETTER MONITORS



JOPPE WOUTERS

dataMinds CONNECT 2026



datasense

October 14th, 2026

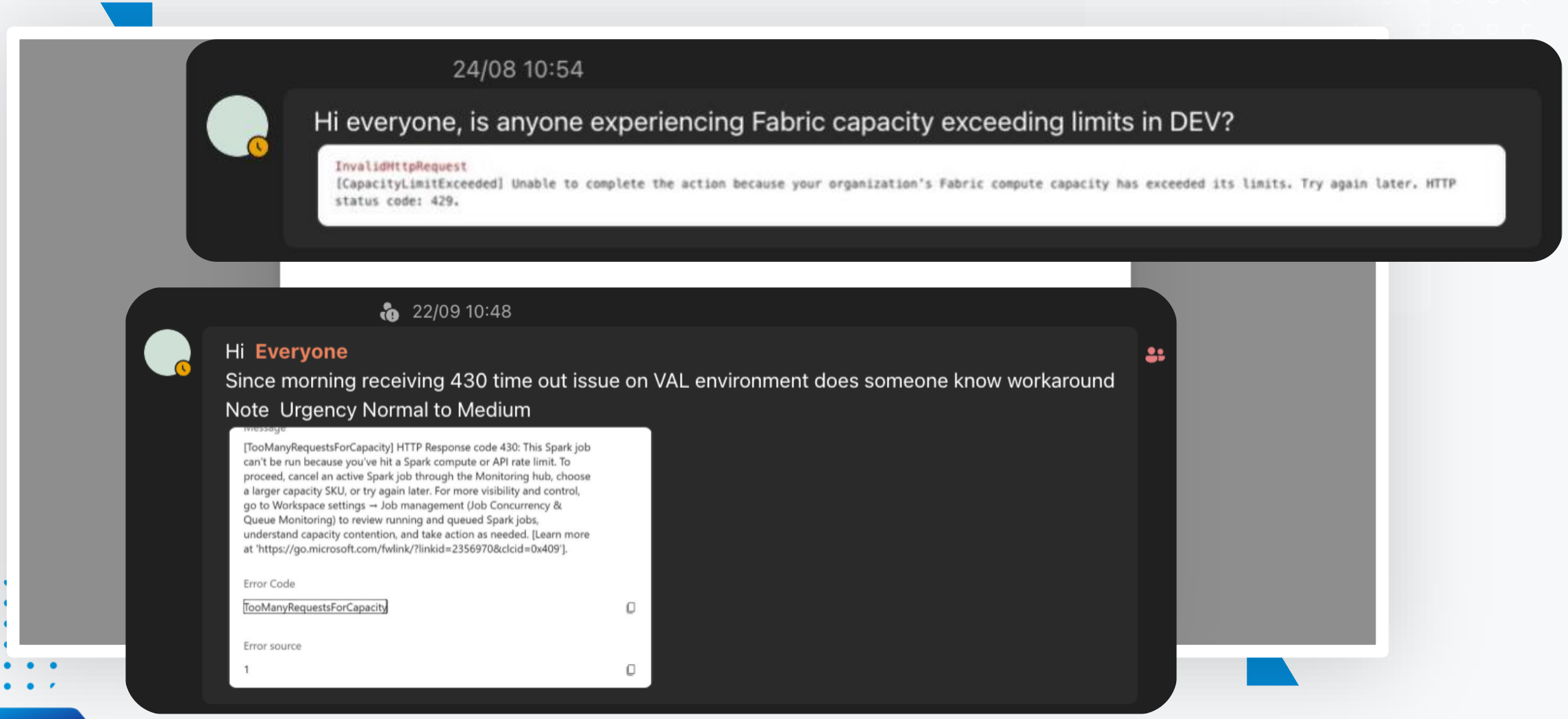
Keeping the Fabric factory operational

Turning cost knobs and building better monitors

[TooManyRequestsForCapacity] This spark job can't be run because you have hit a spark compute or API rate limit

- Microsoft Fabric

© Copyright DataSense



© Copyright DataSense



PRESENTER

Joppe Wouters

Data Engineer at DataSense

© Copyright DataSense





© Copyright DataSense



© Copyright DataSense



© Copyright DataSense



© Copyright DataSense

Fabric cost types

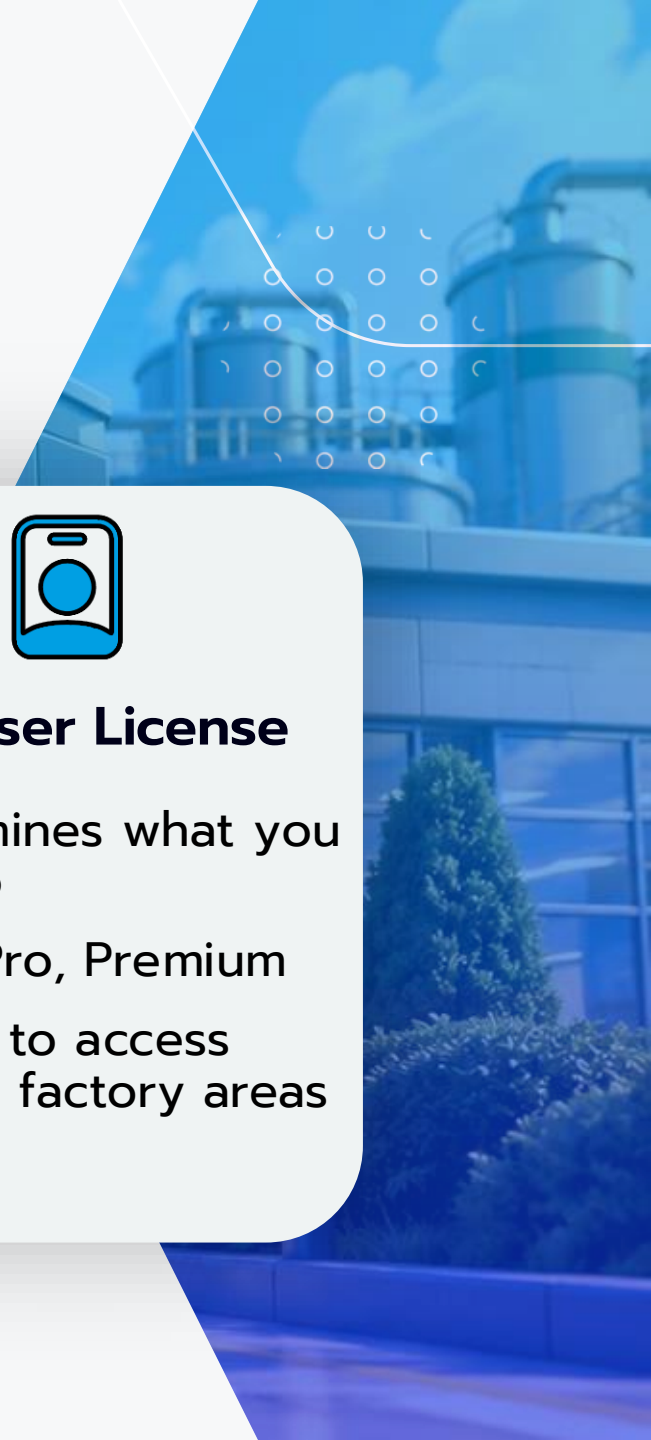


Fabric Capacity

- Provides power for all Fabric workloads
- Sold as Capacity Unit
- Energy sources for the factory

© Copyright DataSense

Fabric cost types



Fabric Capacity

- Provides power for all Fabric workloads
- Sold as Capacity Unit
- Energy sources for the factory

OneLake Storage

- Place to store all your data
- Pay per GB
- Location for factory stock

Per-User License

- Determines what you can do
- Free, Pro, Premium
- Badge to access certain factory areas

© Copyright DataSense



Fabric Capacity

Your factory's power plant

© Copyright DataSense

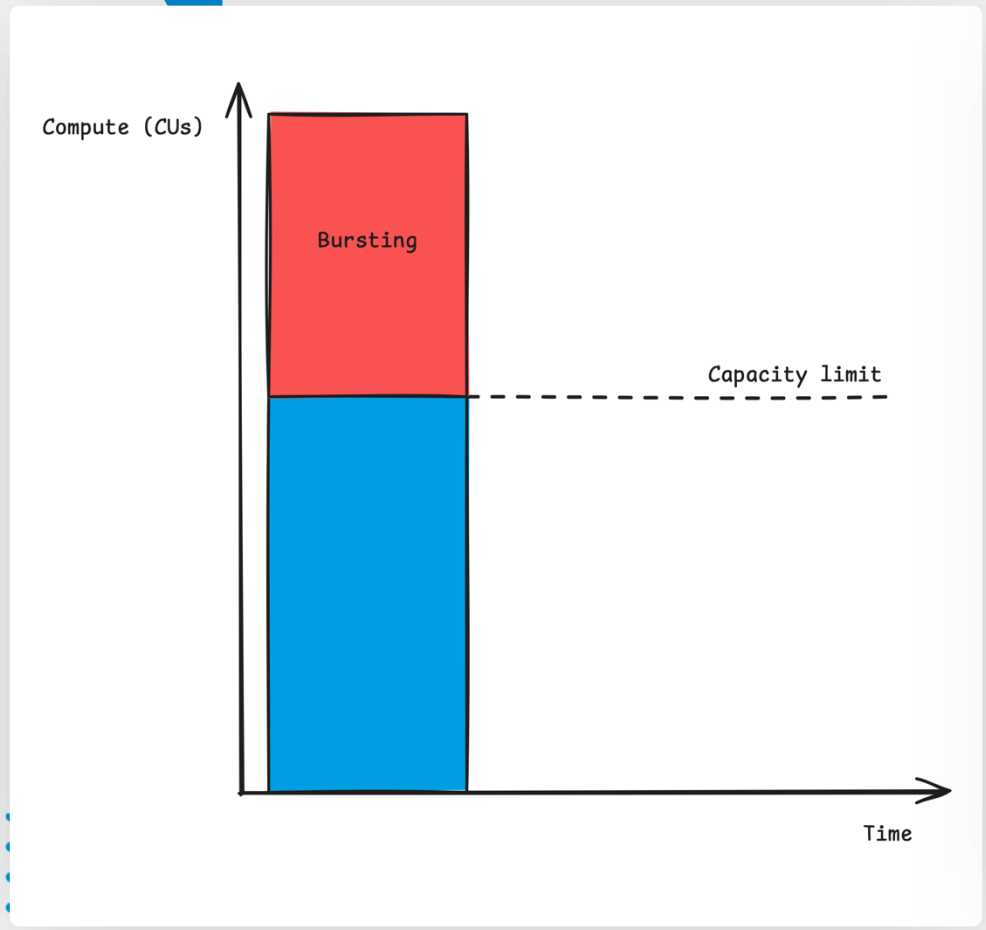


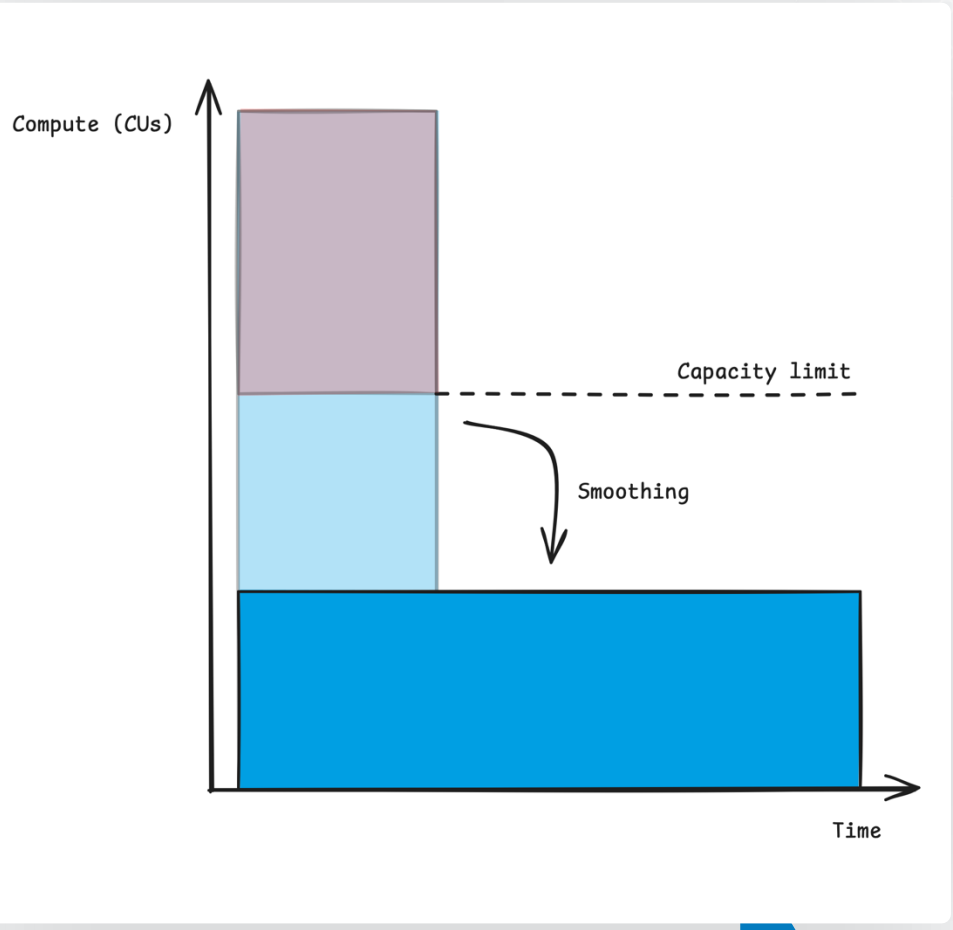
© Copyright DataSense

Fabric capacity sizes

SKU	Capacity Units (CU)	CUs per day
F2	2	172 800
F4	4	345 600
F8	8	691 200
F16	16	1 382 400
...	...	...

© Copyright DataSense





© Copyright DataSense

Fabric operations

Interactive

01 User interactions with the UI

02 Most Power BI operations

03 Smoothing over 5-64 minutes

Background

01 Long running operations and refreshes

02 Data factory pipelines and Notebook interations

03 Smoothing over 24 hours



© Copyright DataSense



© Copyright DataSense

Throttling

01 Too much usage

02 Used up all CUs for the next 10 minutes

03 New operations are rejected upon request



© Copyright DataSense



© Copyright DataSense



© Copyright DataSense

Do I reserve my capacity?

Is it really worth the cost savings?



© Copyright DataSense



© Copyright DataSense



What time do I need it?

© Copyright DataSense

Fabric capacity costs

SKU	Capacity Units (CU)	Pay-as-you-go	Reservation
F2	2	€ 282,647/month	€ 168,075/month ~41% savings
F4	4	€ 565,294/month	€ 336,150/month ~41% savings
F8	8	€ 1 130,588/month	€ 672,299/month ~41% savings
F16	16	€ 2 261,176/month	€ 1 344,597/month ~41% savings
...	...	...	...

© Copyright DataSense

Fabric capacity costs

SKU	Capacity Units (CU)	Pay-as-you-go	Reservation
F2	2	€ 282,647/month	€ 168,075/month ~41% savings
F4	4	€ 565,294/month	€ 336,150/month ~41% savings
F8	8	€ 1 130,588/month	€ 672,299/month ~41% savings
F16	16	€ 2 261,176/month	€ 1 344,597/month ~41% savings
...	...	...	...

© Copyright DataSense

The 59% usage rule

- Reservation cost benefit appears after 59% usage

	A	B	C	D	E	F	G	H	I	J
1	SKU	Days	Start hour	End hour	Totaal hours	Capacity on	PAYG 24/7	Reservation	PAYG CPR	Monthly savings*
2	F2	5	7	19	60	0.357142857	€ 282.65	€ 168.07	€ 100.95	€ 67.13
3	F4	5	7	19	60	0.357142857	€ 565.29	€ 336.15	€ 201.89	€ 134.26
4	F8	5	7	19	60	0.357142857	€ 1,130.59	€ 672.30	€ 403.78	€ 268.52
5	F16	5	7	19	60	0.357142857	€ 2,261.18	€ 1,344.60	€ 807.56	€ 537.03

© Copyright DataSense



© Copyright DataSense



© Copyright DataSense



© Copyright DataSense

The hybrid approach

01 Combining a capacity with consumption-based

02 Consumption-based pricing uses PAYG rate*

03 Less predictable in terms of costs

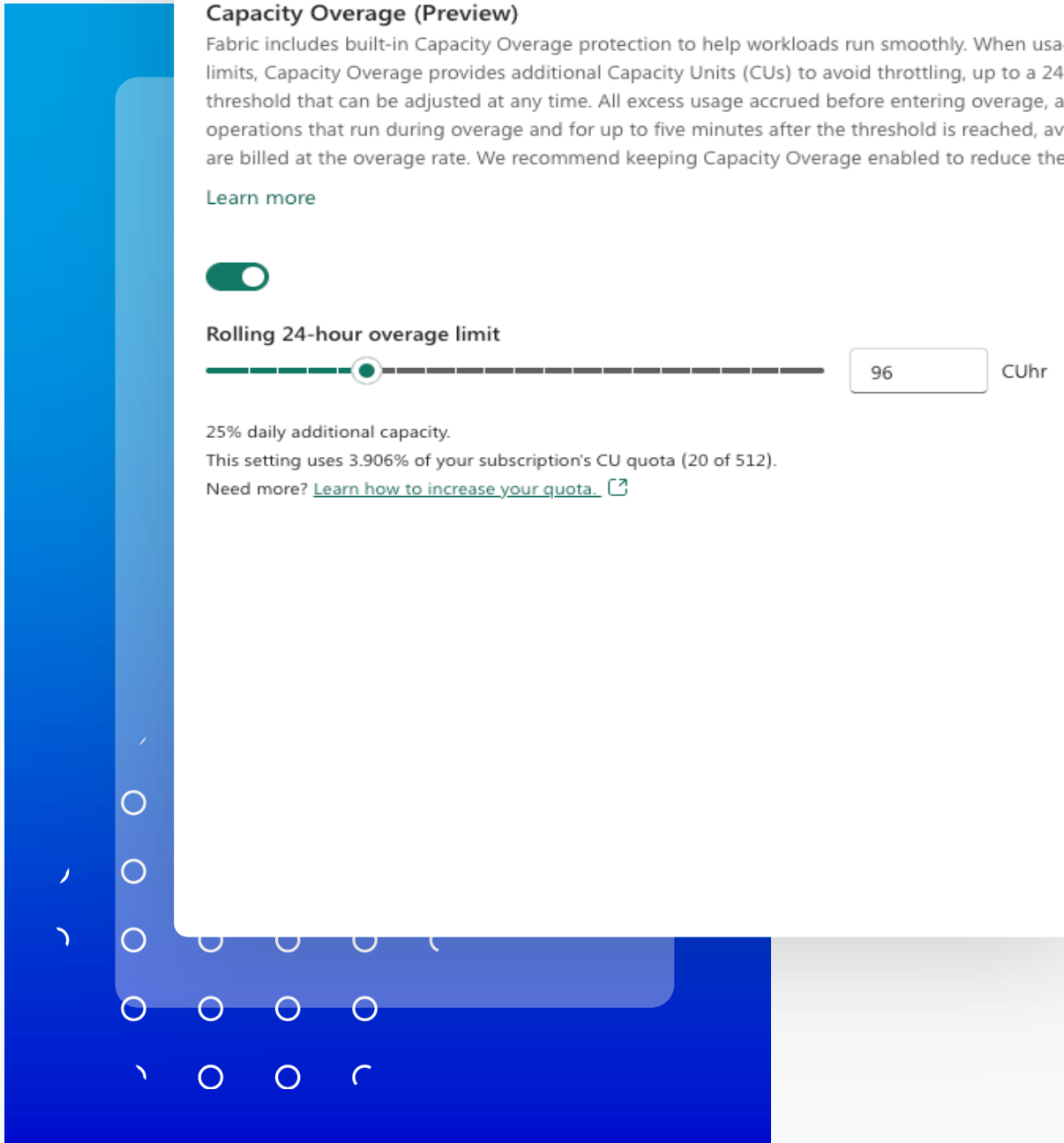
04 Cost guardrails available



© Copyright DataSense

Capacity overage

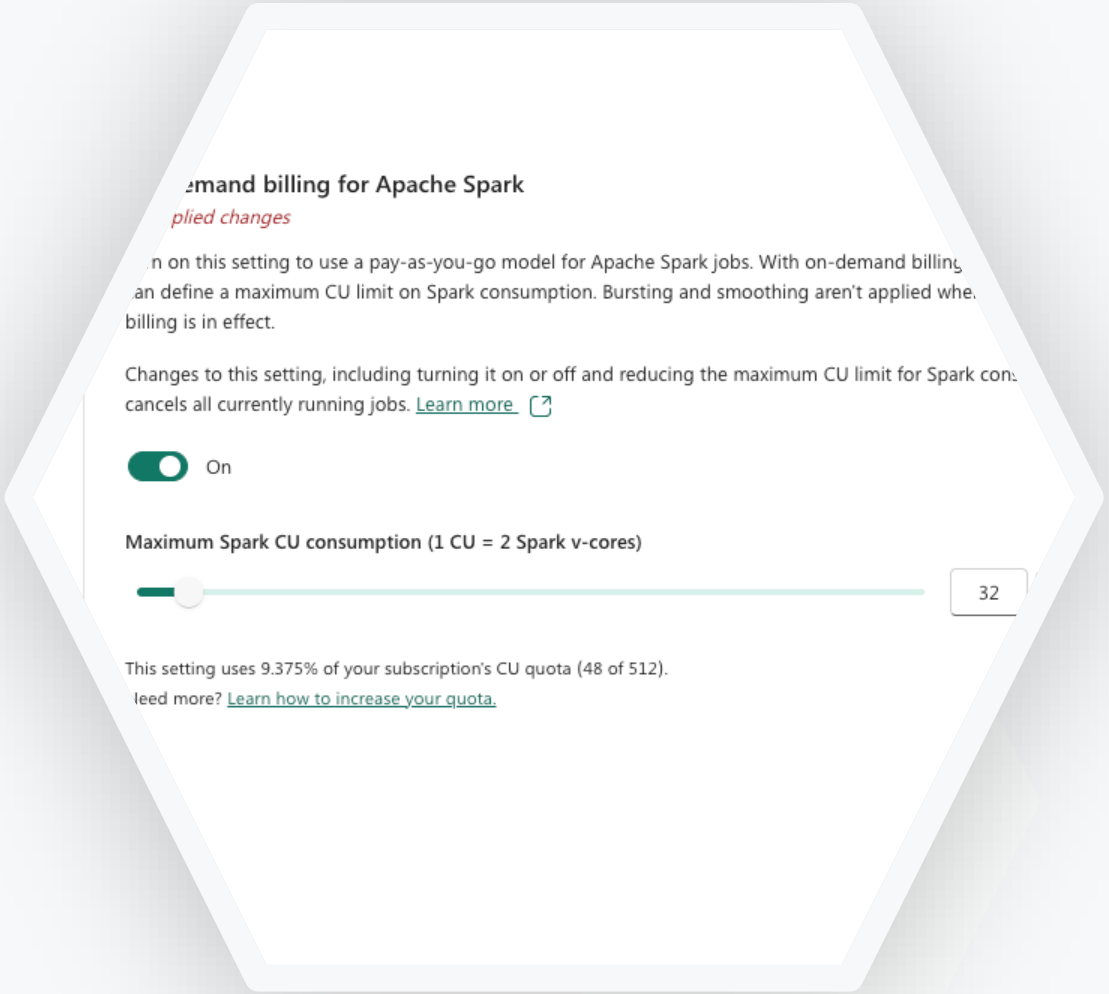
GA since FabCon Barcelona



© Copyright DataSense

On-demand billing for Apache Spark

Previously known as *Autoscale billing for Spark*



© Copyright DataSense

On-demand billing for Apache Spark

01 CU setting dictates concurrency & queue

02 No more bursting or smoothing

03 Scheduling nighly runs don't need fine-tuning



© Copyright DataSense

Latest announcements

On-demand billing

- Based on Spark concept
- Enable per category
- Category usage limit
- Capacity setting



© Copyright DataSense

Latest announcements

On-demand billing

- Based on Spark concept
- Enable per category
- Category usage limit
- Capacity setting

The F0 capacity

- Full consumption-based capacity
- Price multipliers
- 1x to 3x
- Experiments or POCs



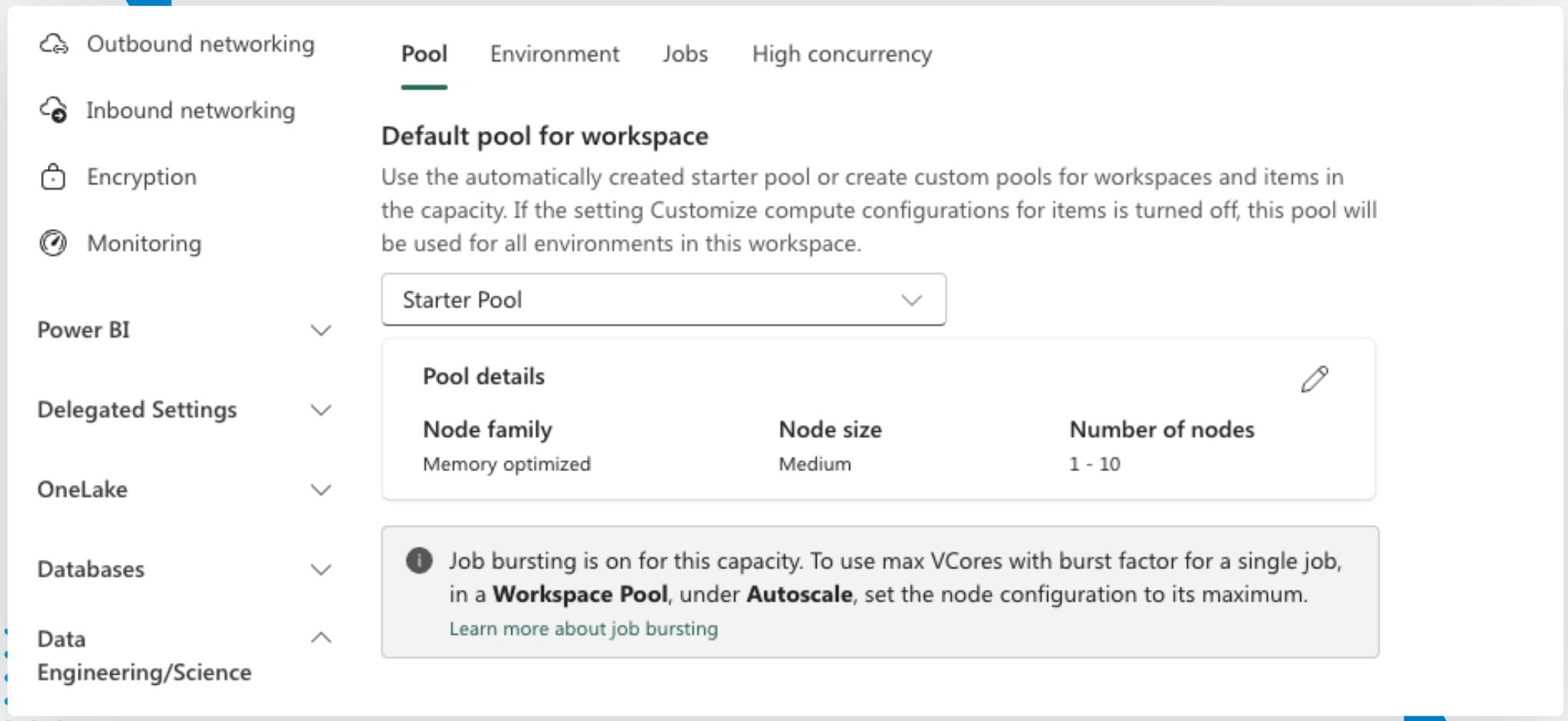
© Copyright DataSense

Spark pools



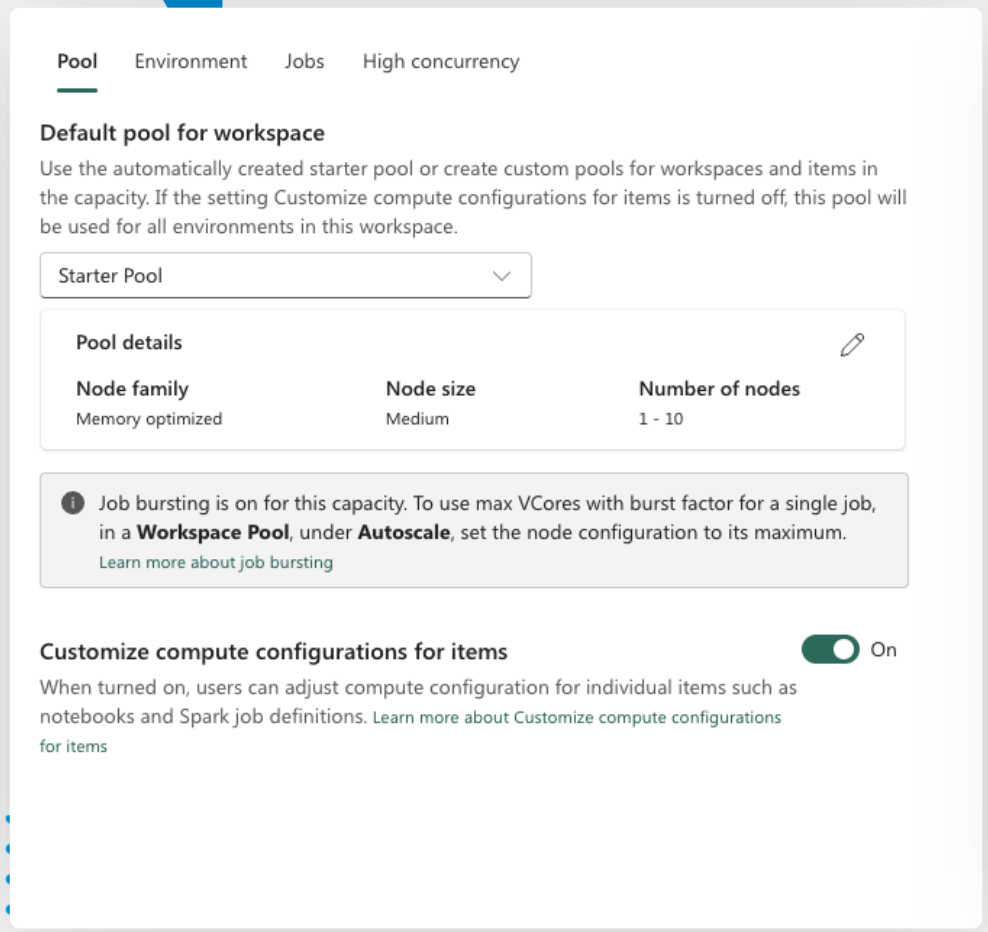
© Copyright DataSense

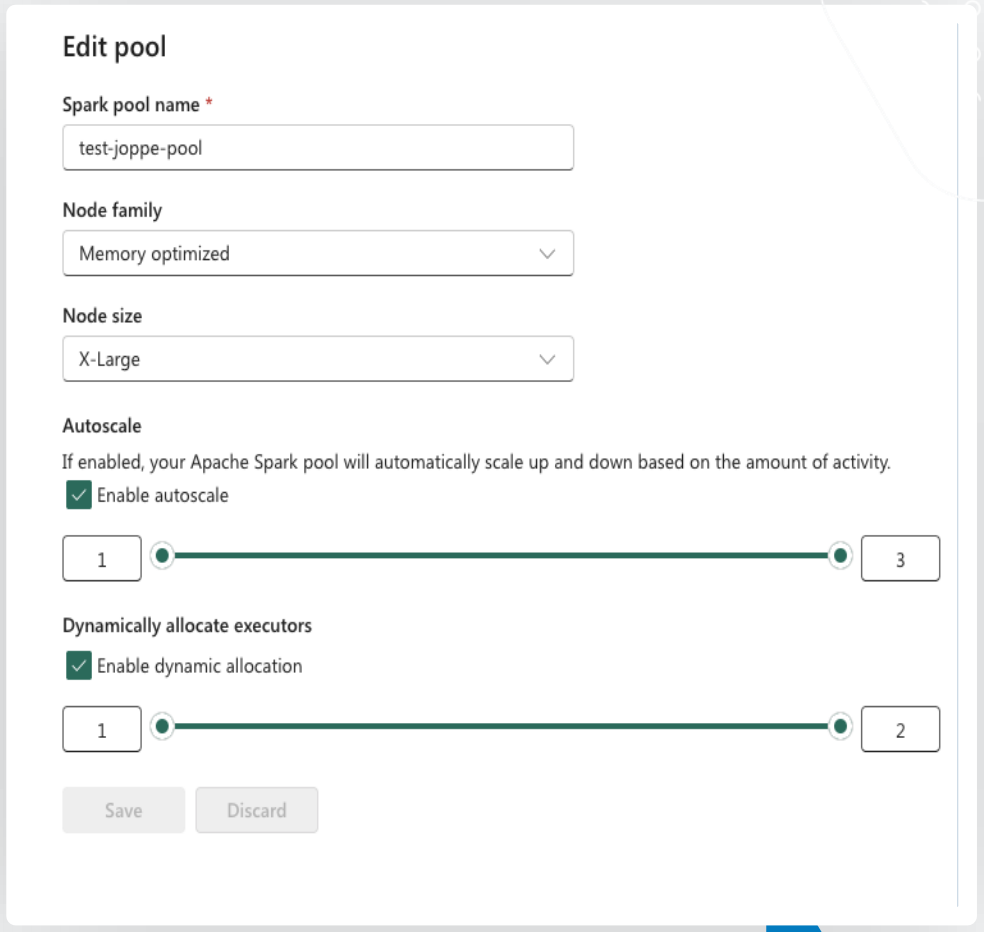
Spark pool settings



© Copyright DataSense

Spark pool settings





© Copyright DataSense

Spark node sizes

Size	vCores	Memory
Small	4	32 GB
Medium	8	64 GB
Large	16	128 GB
X-Large	32	256 GB
XX-Large	64	512 GB

© Copyright DataSense

Spark node sizes

Size	vCores	Memory
Small	4	32 GB
Medium	8	64 GB
Large	16	128 GB
X-Large	32	256 GB
XX-Large	64	512 GB

© Copyright DataSense

Fabric capacity vCores

SKU	Capacity Units (CU)	vCores	vCores with burst (3x)
F2	2	4	12
F4	4	8	24
F8	8	16	48
F16	16	32	96
...	...	...	...

© Copyright DataSense

Spark on F4

Default settings

01 Starter pool

02 Medium node size with autoscale enabled

03 8 vCores for driver and 8 vCores for worker

04 1 active notebook session

Single-node Spark pool

01 Only disable autoscale

02 8 vCores

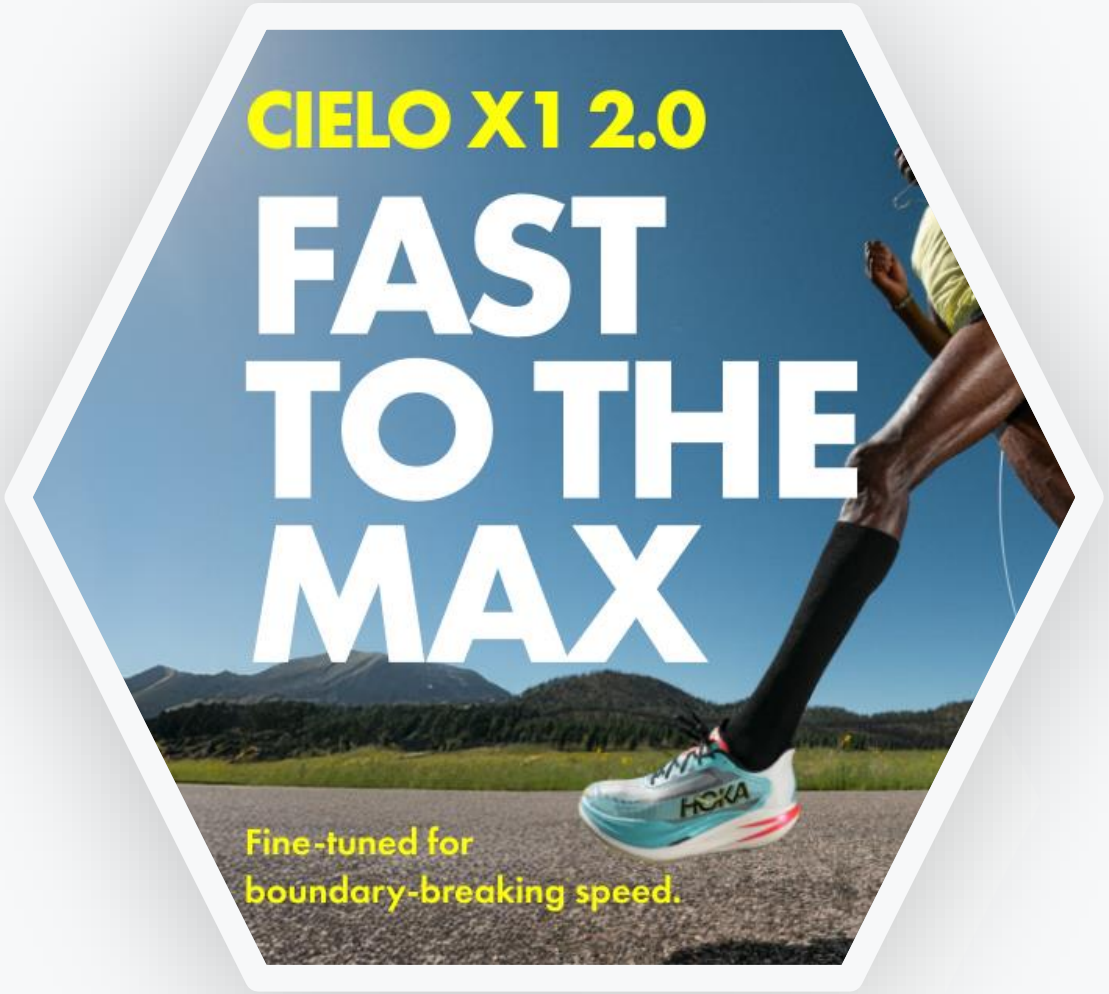
03 Up to 3 active notebook sessions

04 Perfect for development workspaces



© Copyright DataSense

Runtime 2.0



© Copyright DataSense

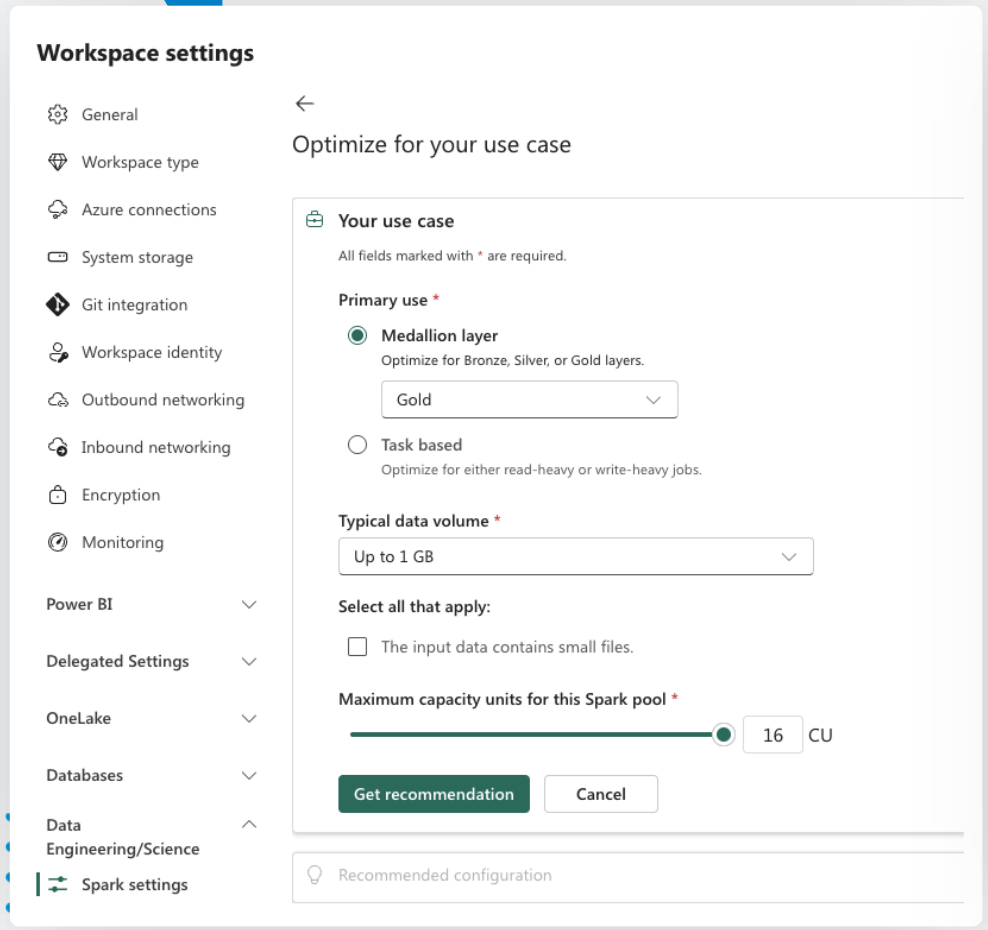
Native Execution Engine

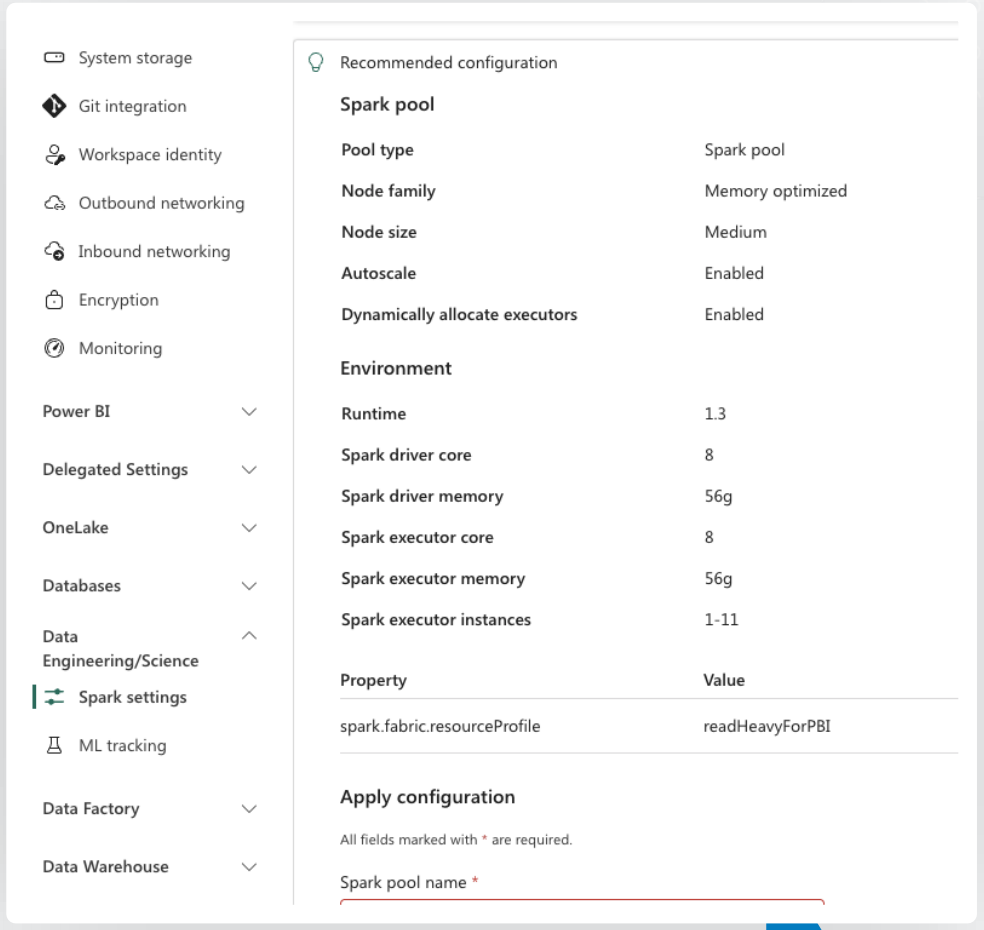
spark.conf.set('spark.native.enabled', 'true')



© Copyright DataSense

Tuning is too technical





© Copyright DataSense

Spark notebooks in pipelines



© Copyright DataSense

Orchestration methods



© Copyright DataSense



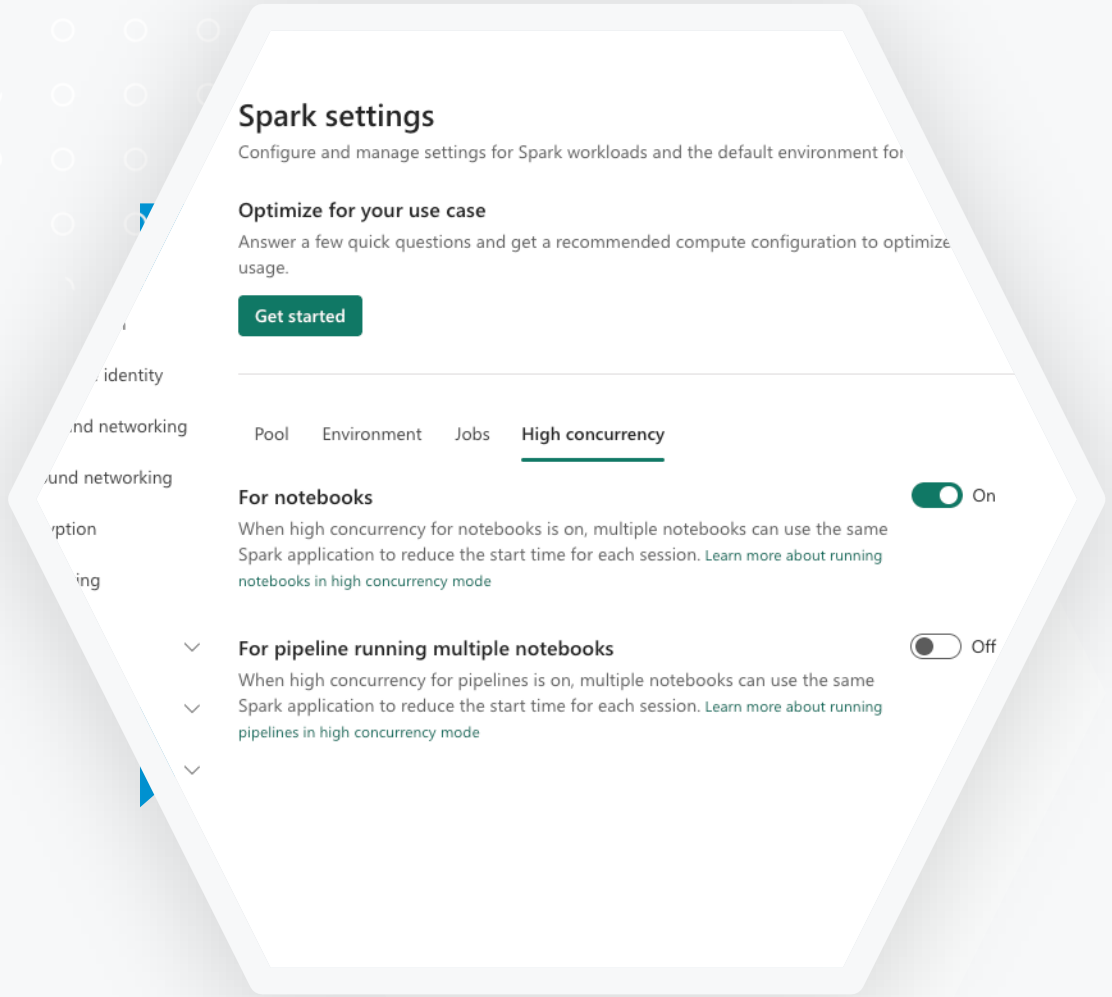
© Copyright DataSense



© Copyright DataSense

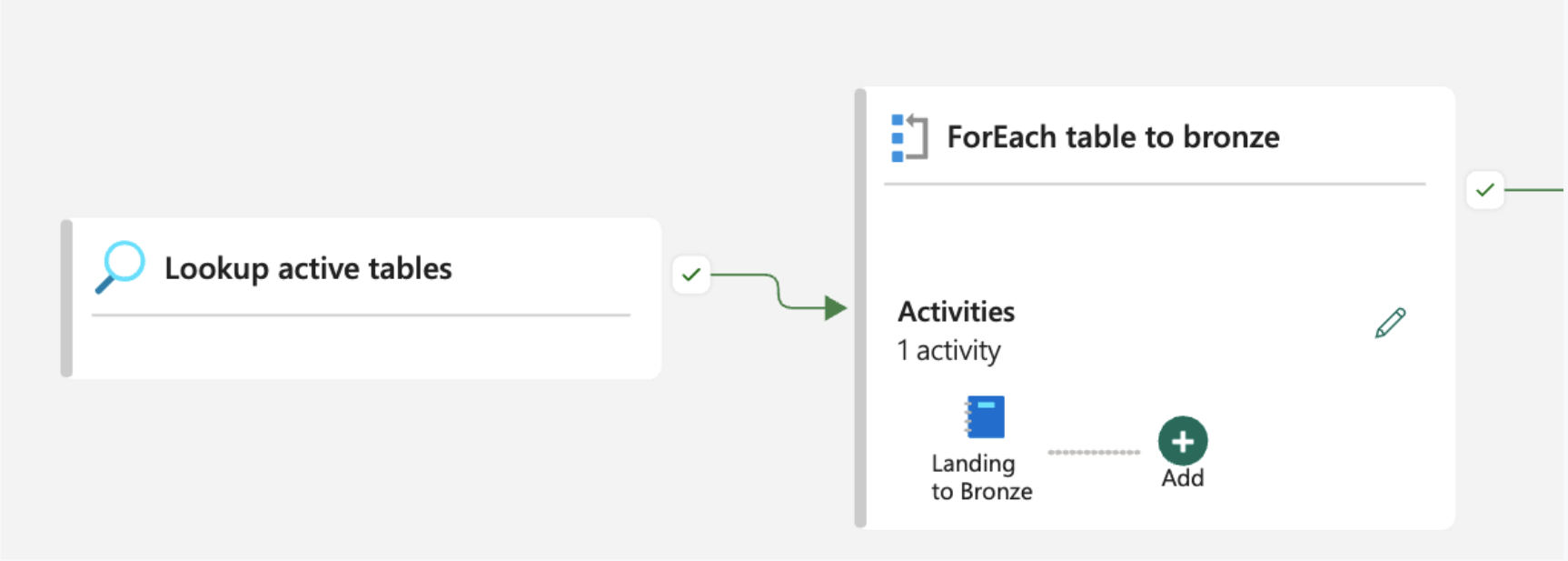
High concurrency

The opt-in setting that should be enabled by default



© Copyright DataSense

Notebooks in pipelines





© Copyright DataSense

Sharing HC sessions

On Fabric don’t be like Joey and share!

01 Default lakehouse in notebook

02 Session tags in pipeline activity

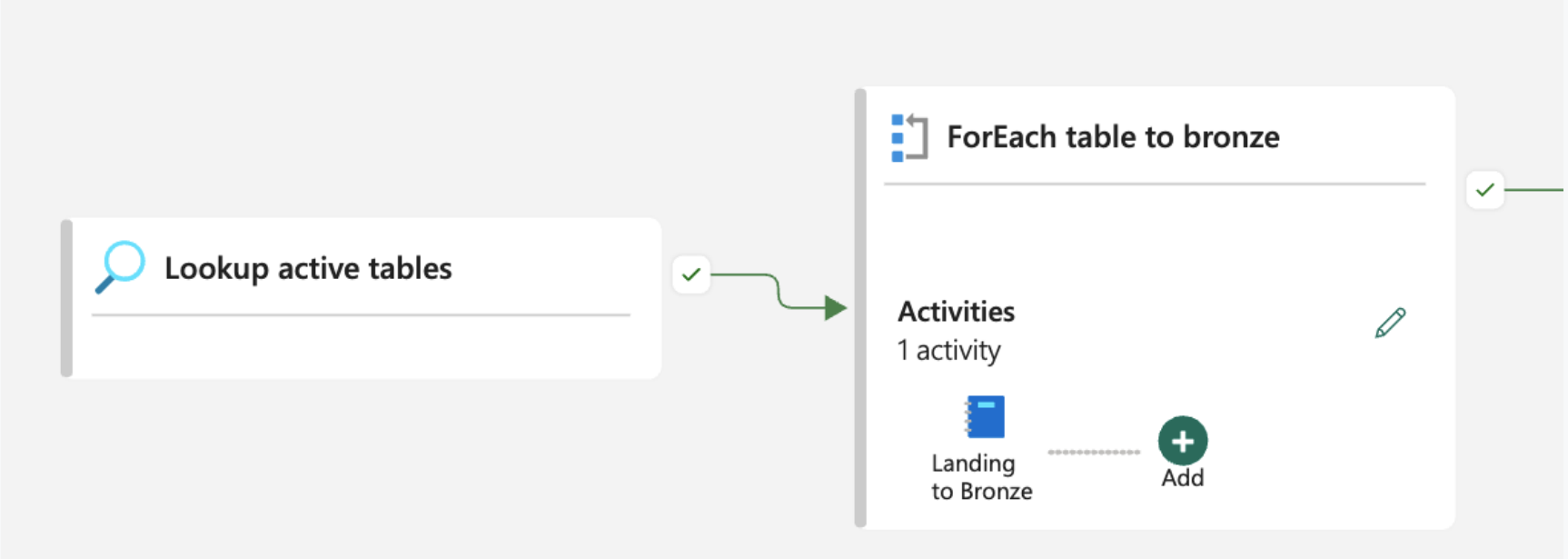
03 Maximum 5 notebooks in 1 HC session





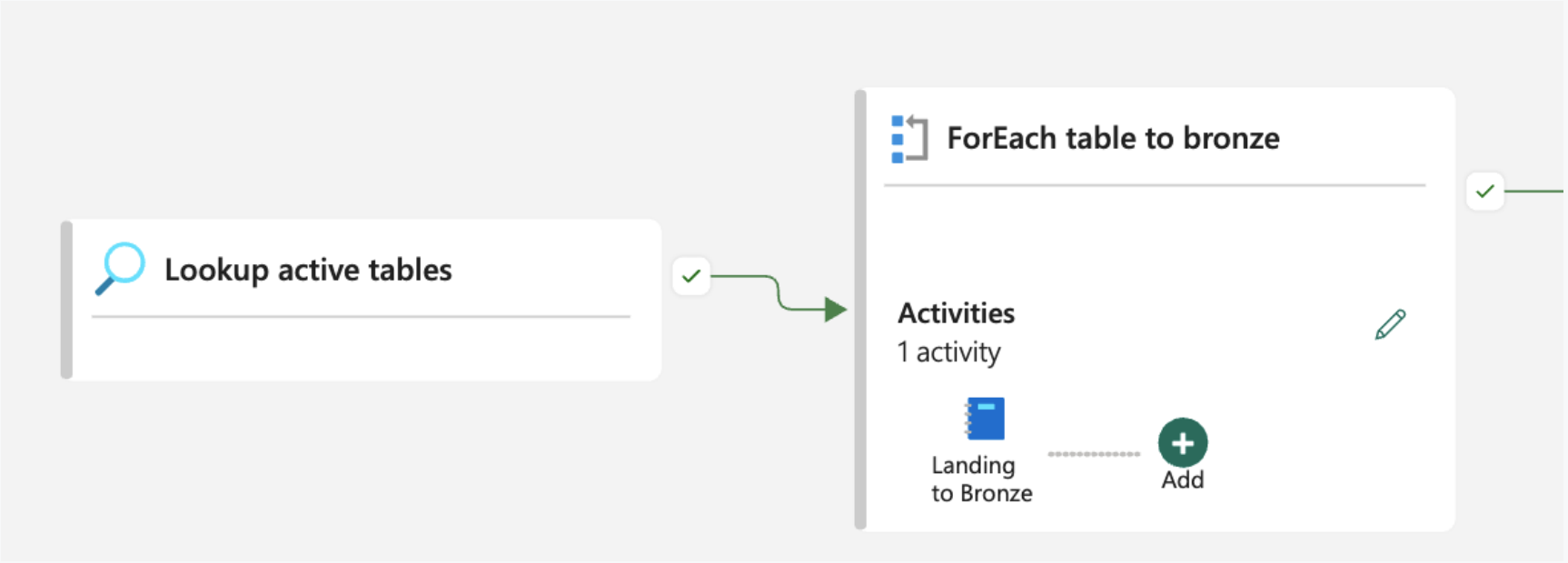
© Copyright DataSense

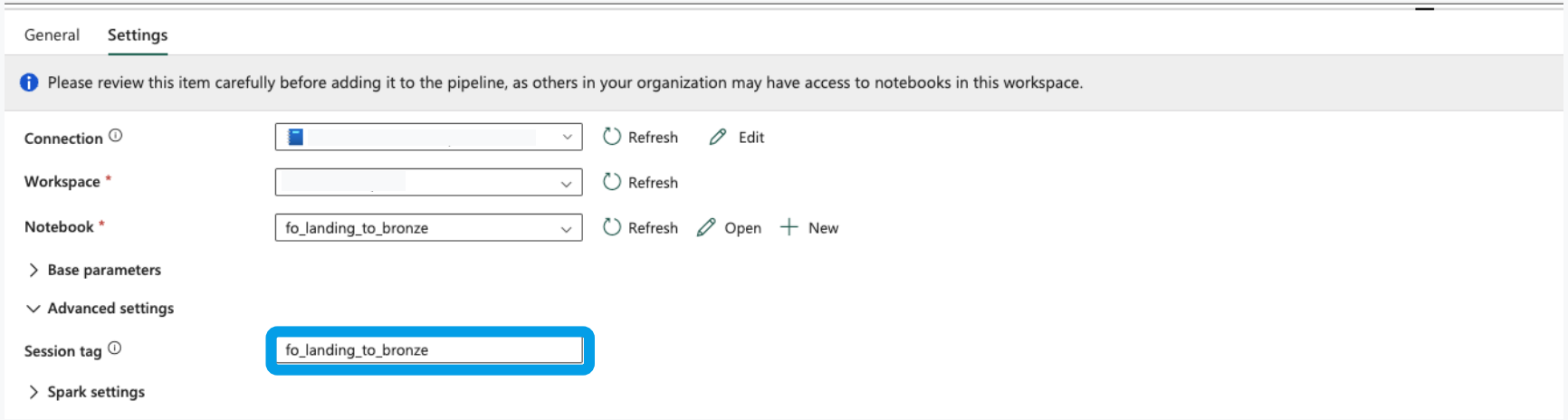
Notebooks in pipelines



© Copyright DataSense

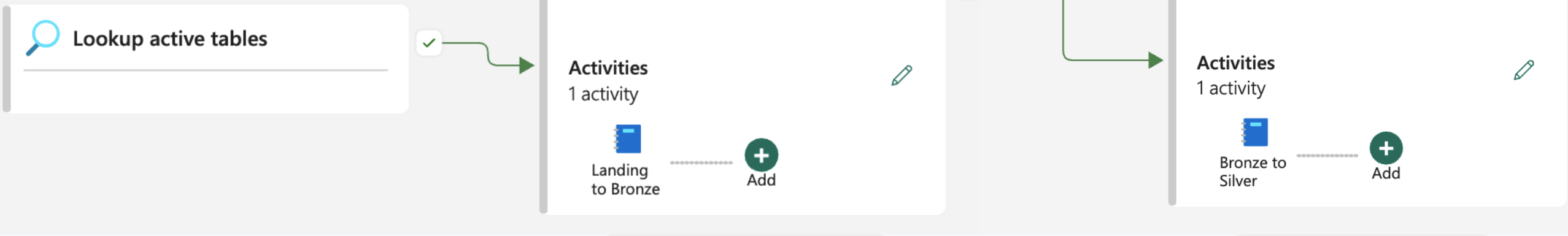
Notebooks in pipelines

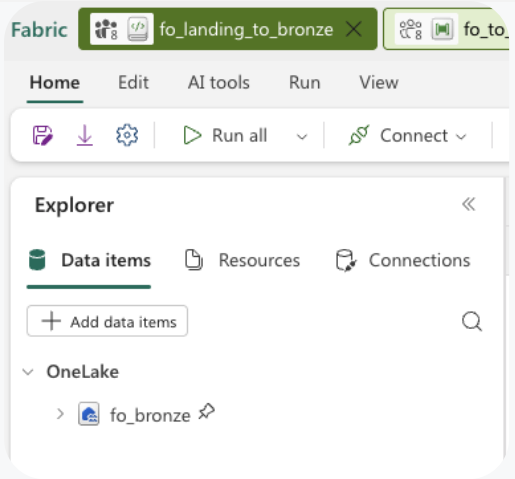


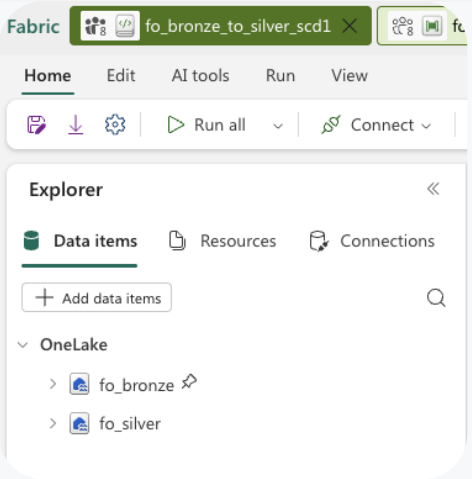


© Copyright DataSense

Notebooks in pipelines







© Copyright DataSense



© Copyright DataSense

Orchestrator notebook

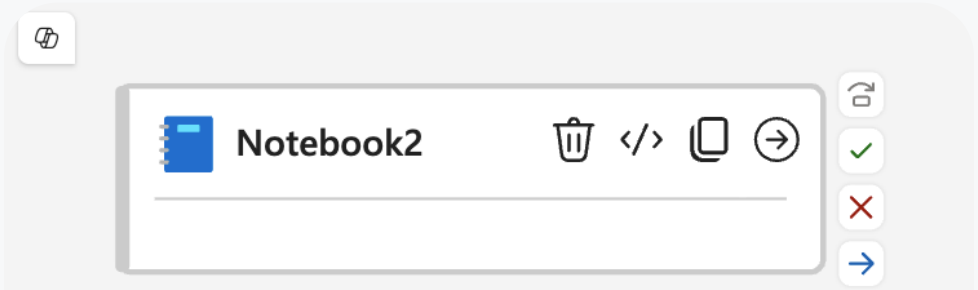
notebookutils.notebook.runMultiple()

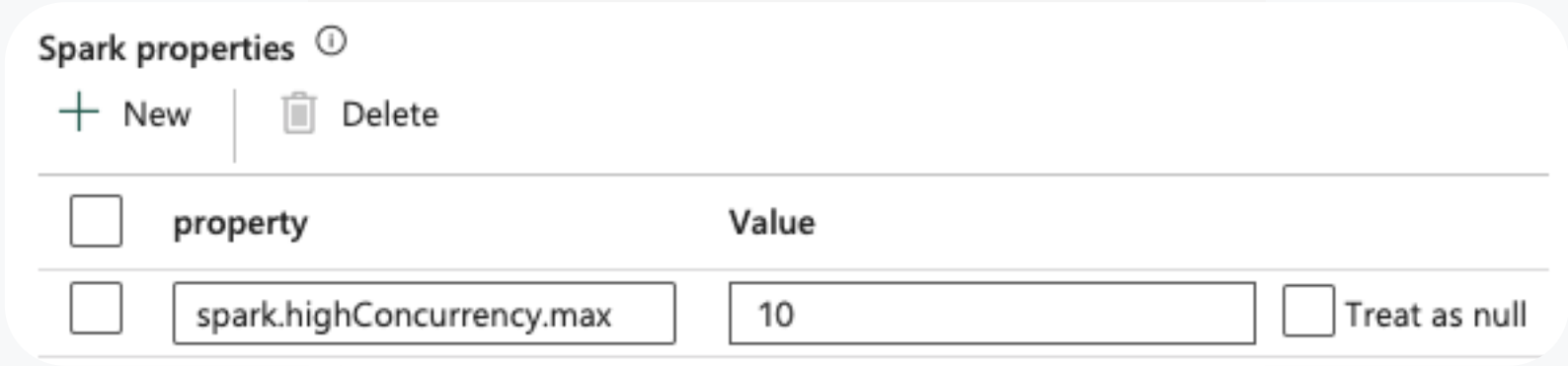
- 1 Spark session
- Maximum 50 notebooks
- DAG structure
- Queuing system
- Metadata lookup

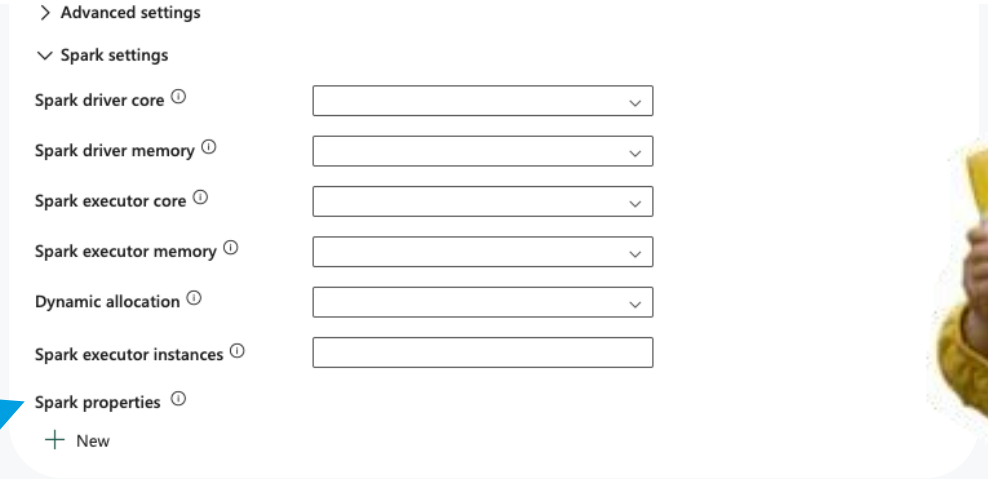


© Copyright DataSense

Notebook Spark settings









© Copyright DataSense

Where do you begin?

01 Enable high concurrency on your workspace

02 Choose your the default lakehouse for HC

03 Set the HC Spark property

04 Orchestrator with DAG

© Copyright DataSense



© Copyright DataSense

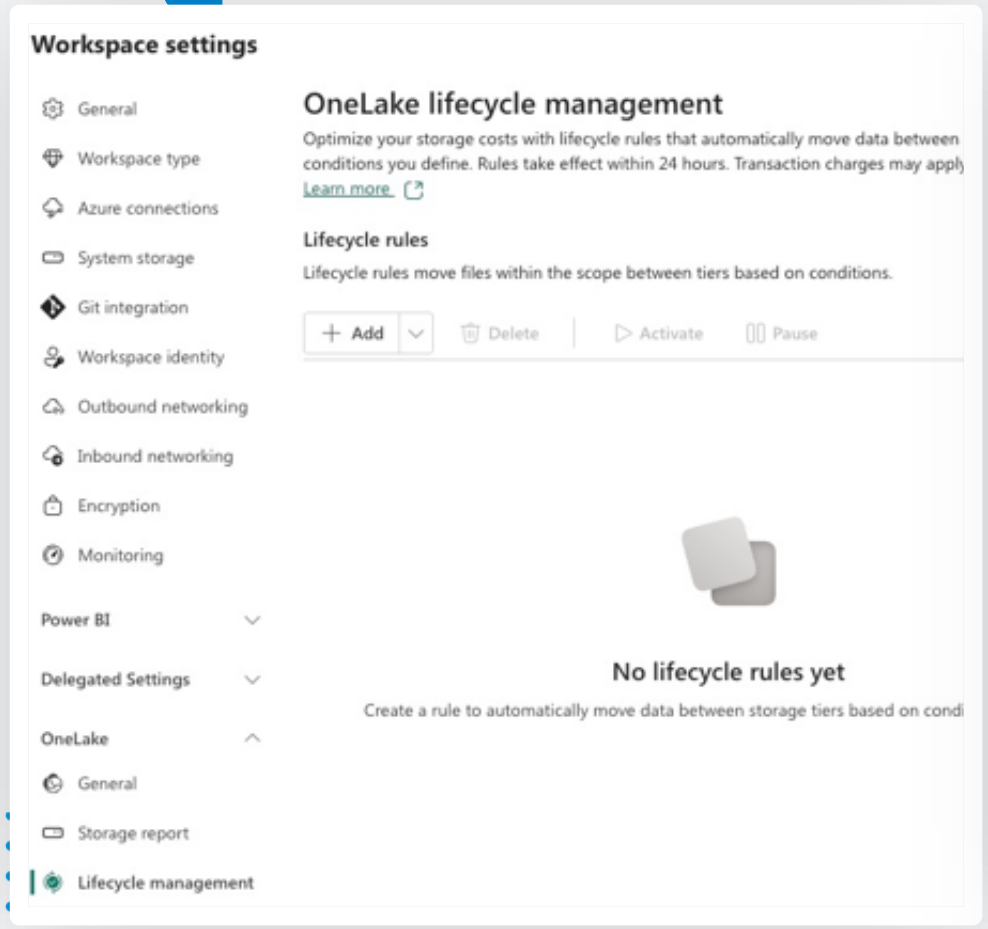


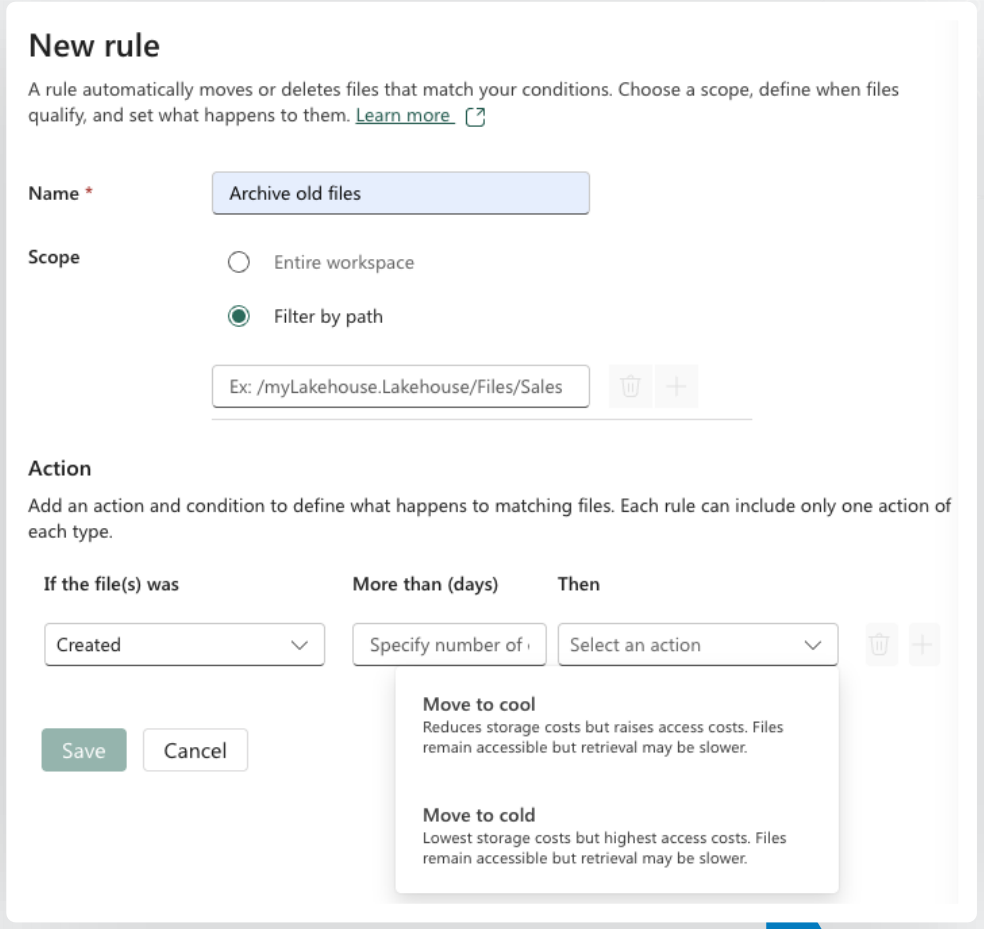
© Copyright DataSense



© Copyright DataSense

Lifecycle management





© Copyright DataSense

Maintain your Delta tables

01 Check your workspace's storage report

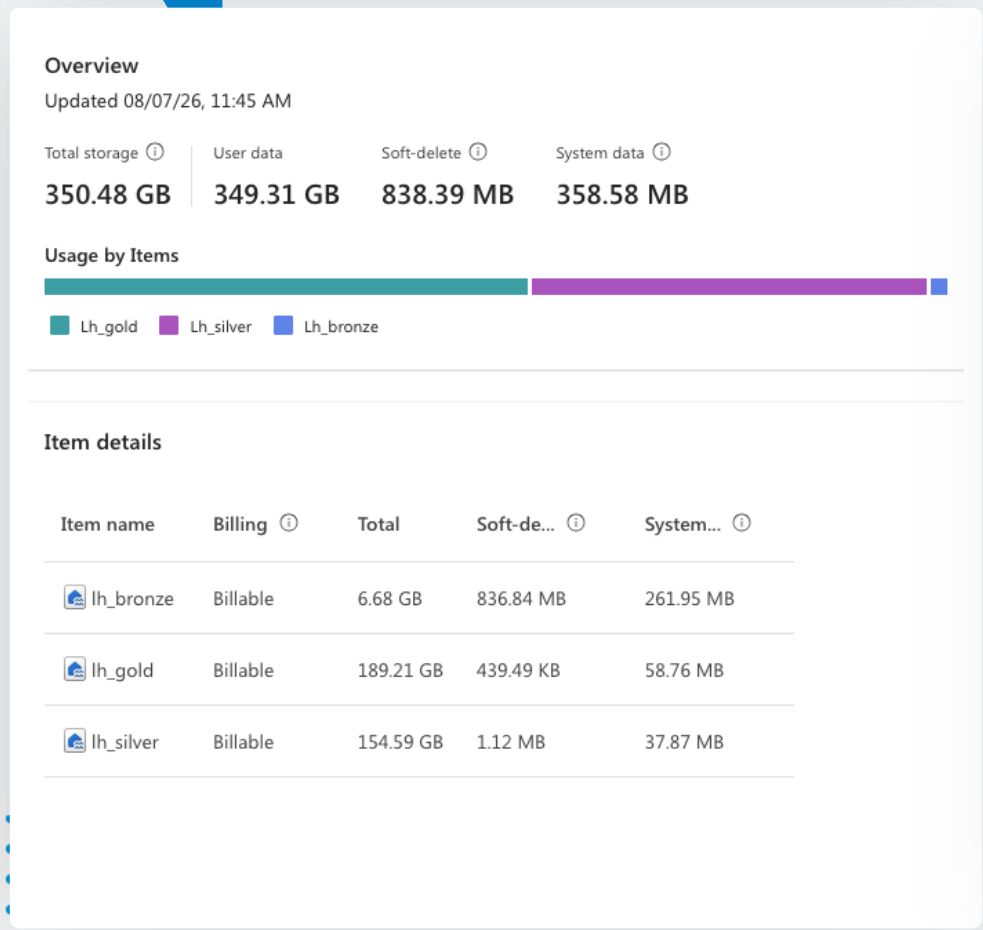
02 Run commands like VACUUM & OPTIMIZE

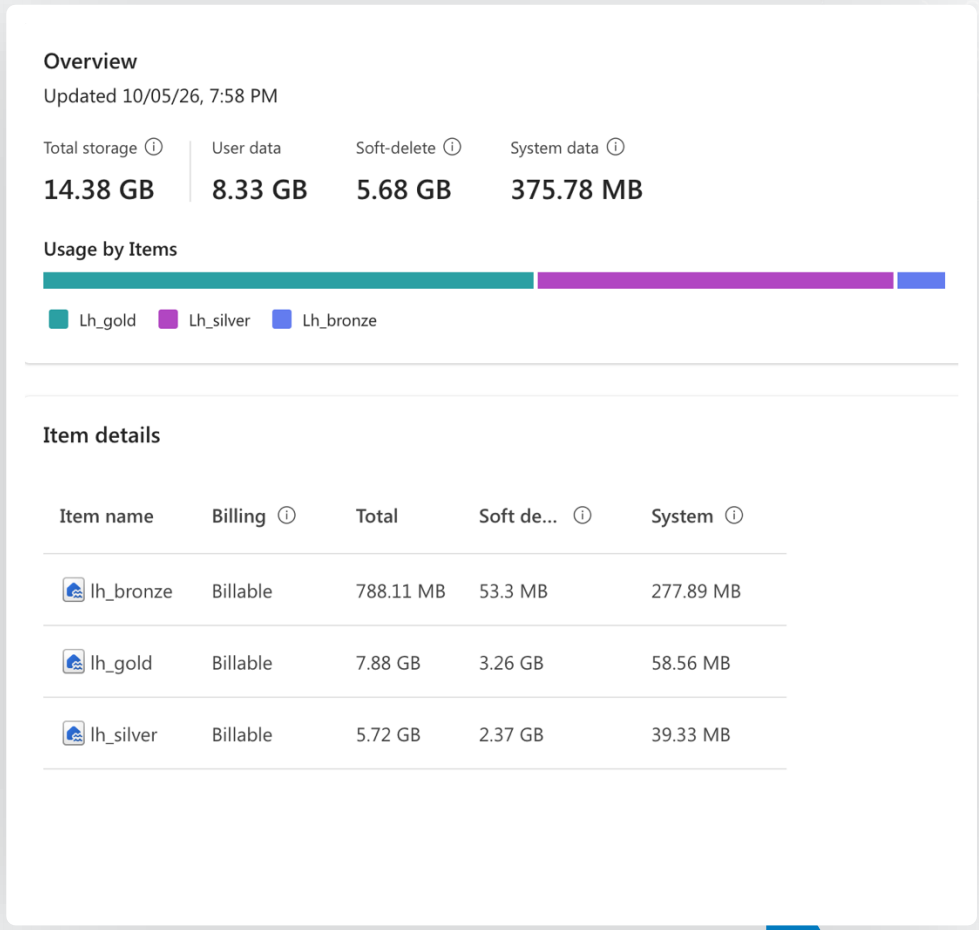
03 Schedule a maintenance notebook



© Copyright DataSense

Why VACUUM matters

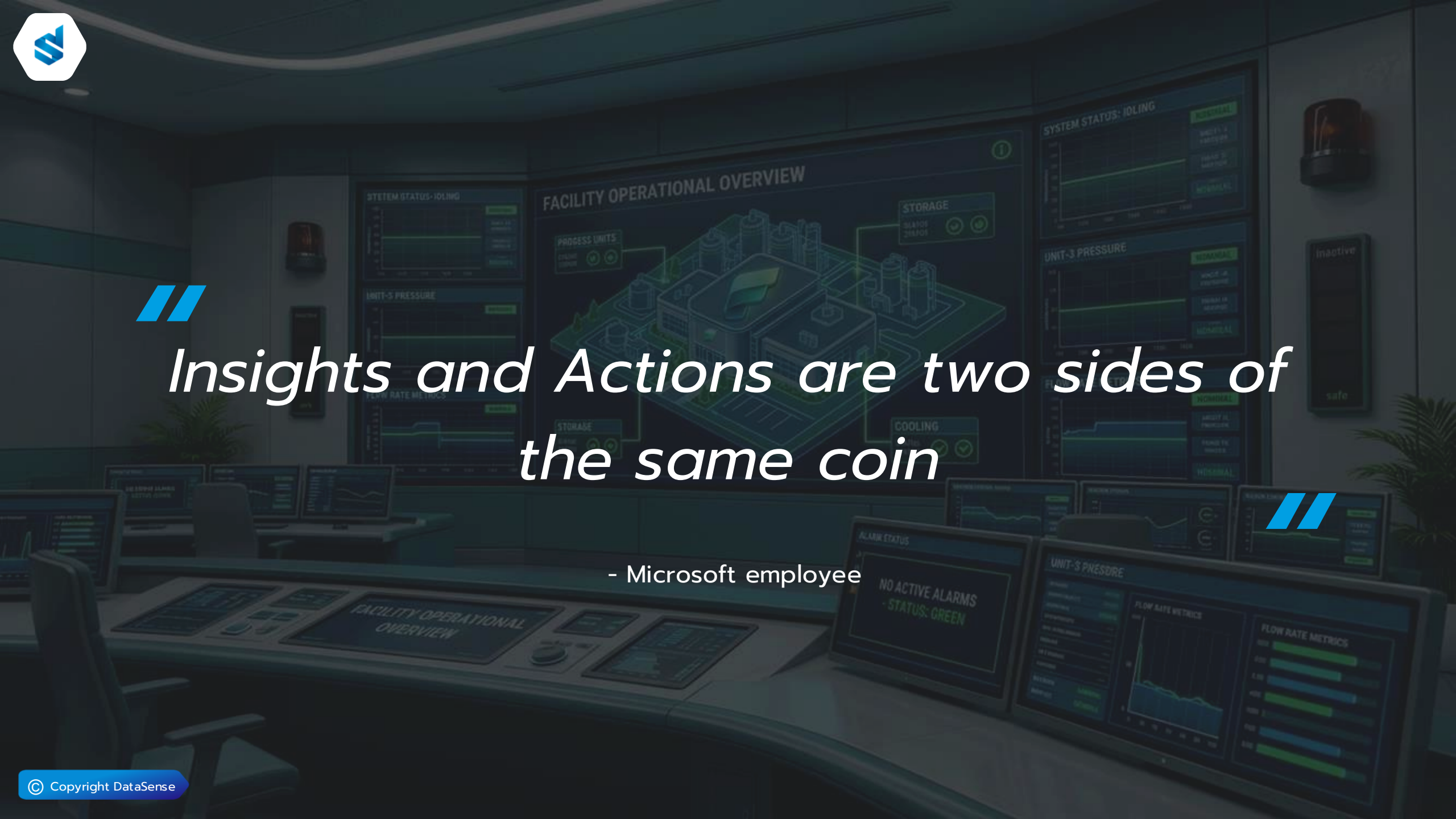




© Copyright DataSense



© Copyright DataSense



"Insights and Actions are two sides of the same coin"

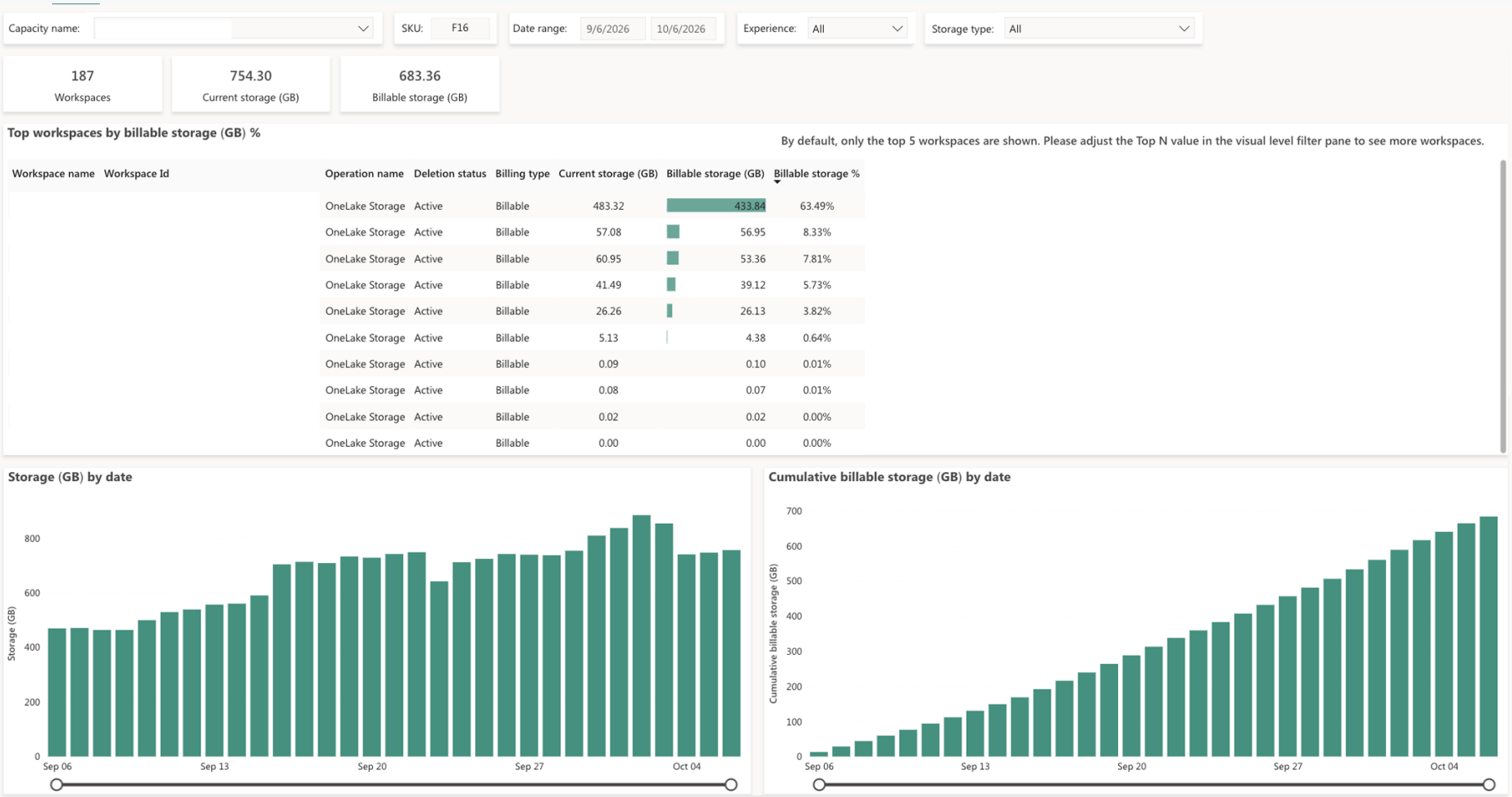
- Microsoft employee

© Copyright DataSense



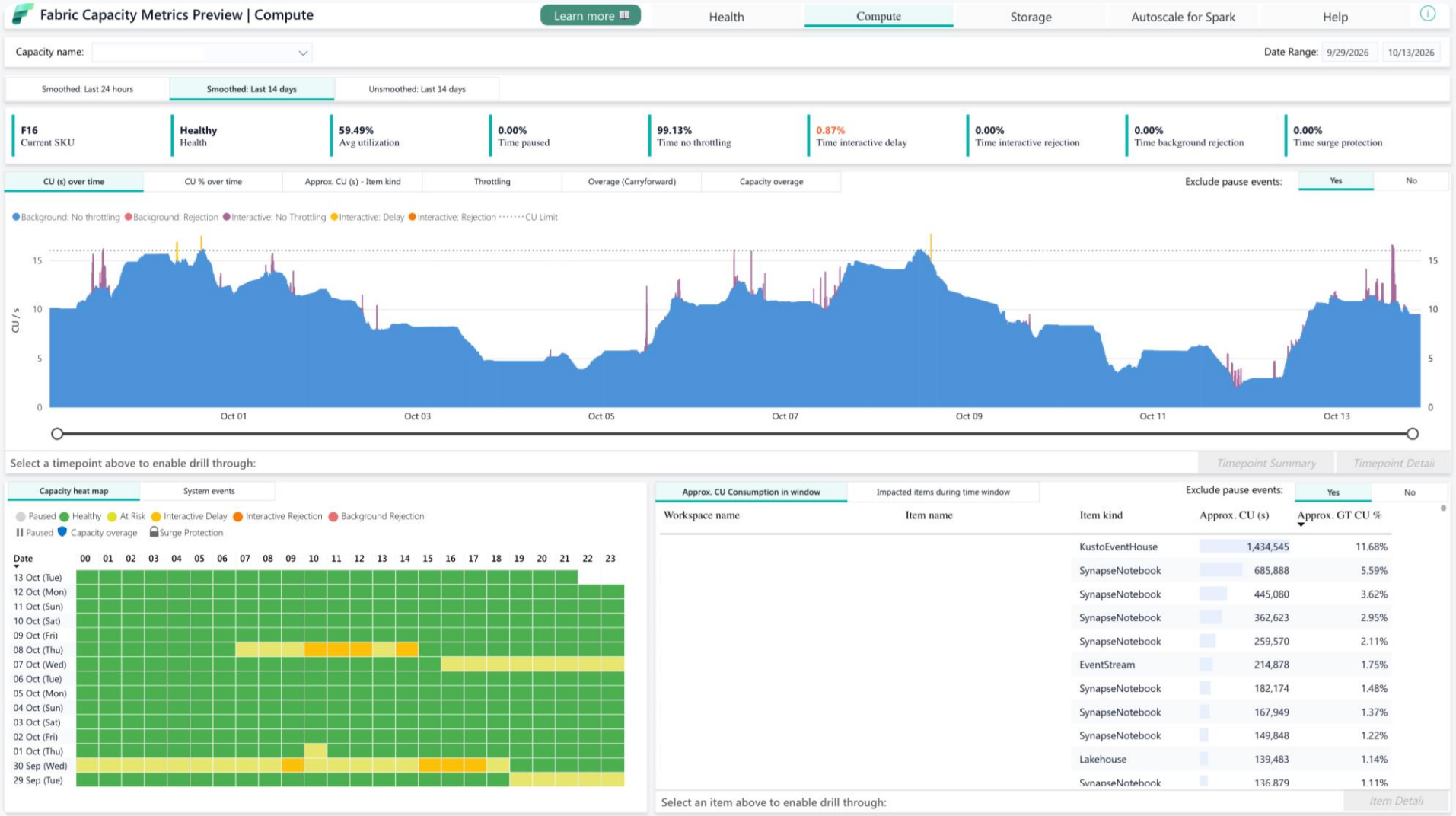
© Copyright DataSense

Capacity Metrics



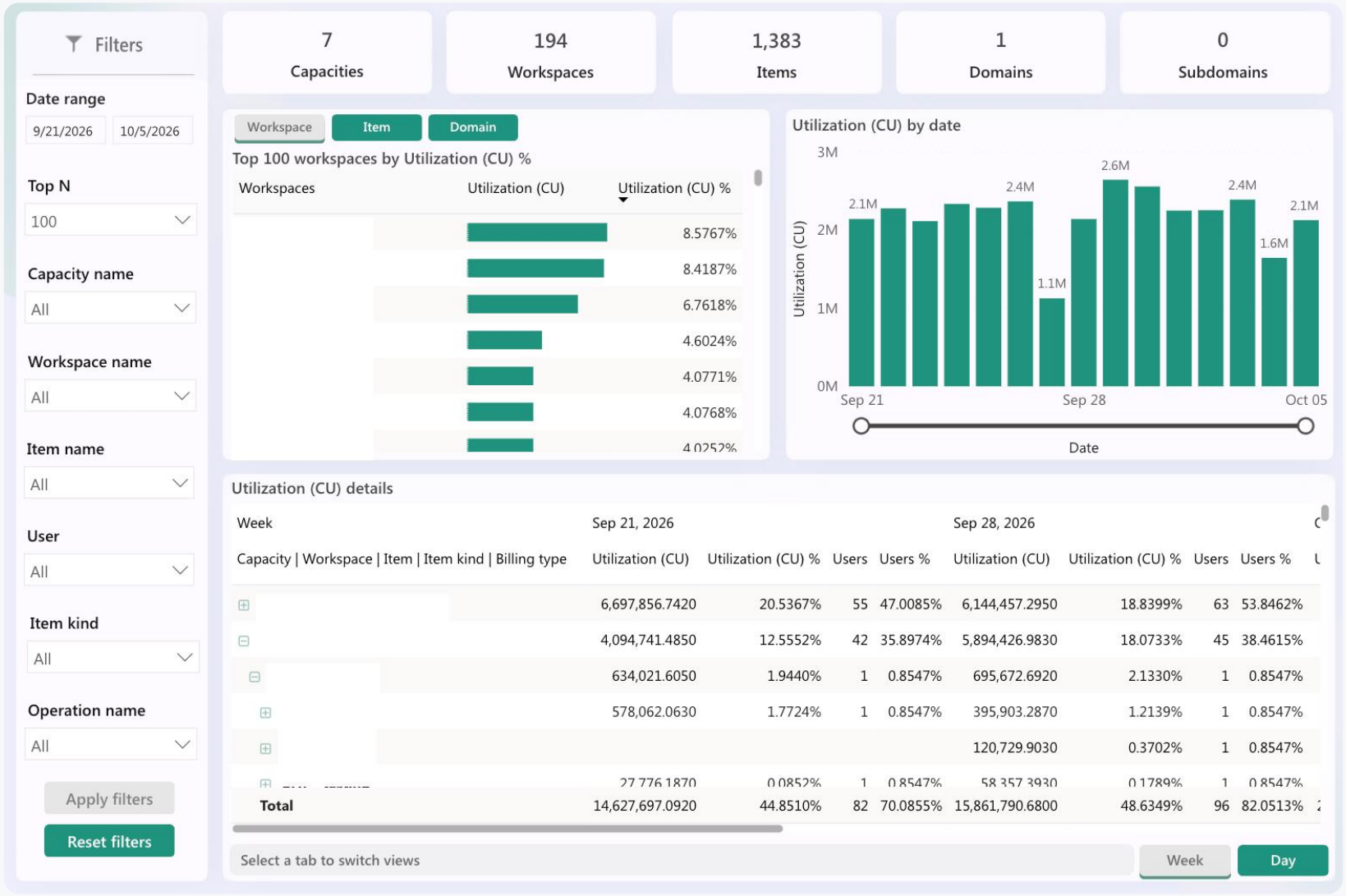
© Copyright DataSense

Capacity Metrics (preview)



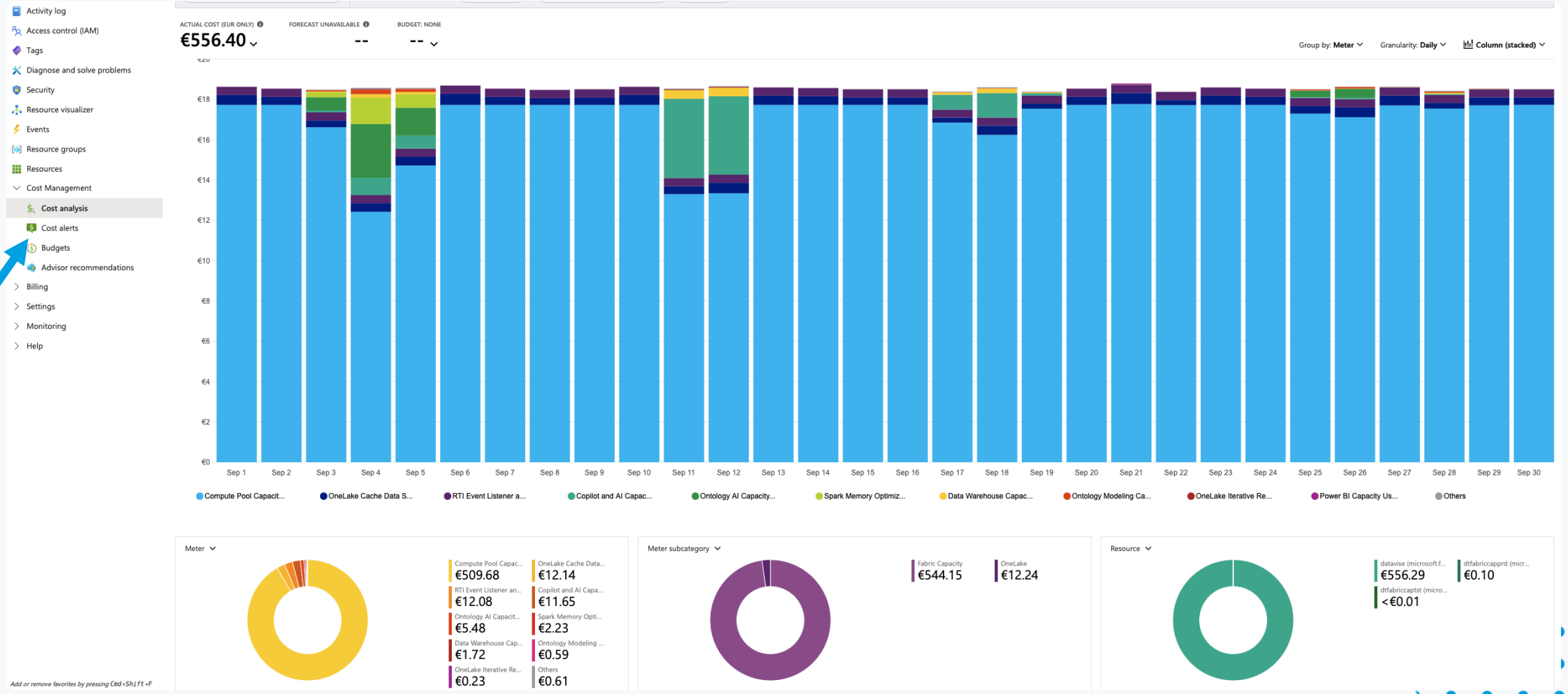
© Copyright DataSense

Chargeback Reporting



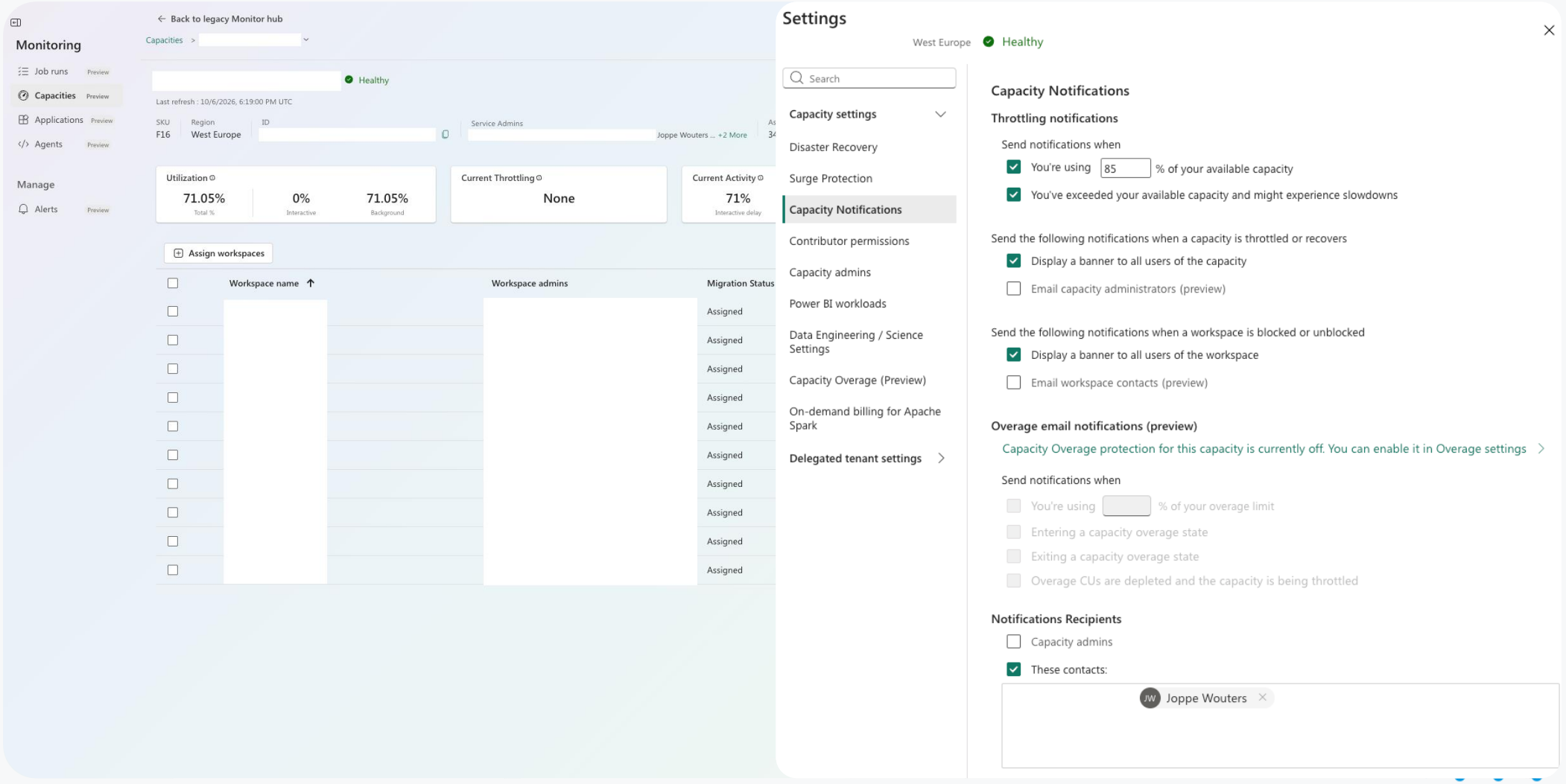
© Copyright DataSense

Azure Cost analysis



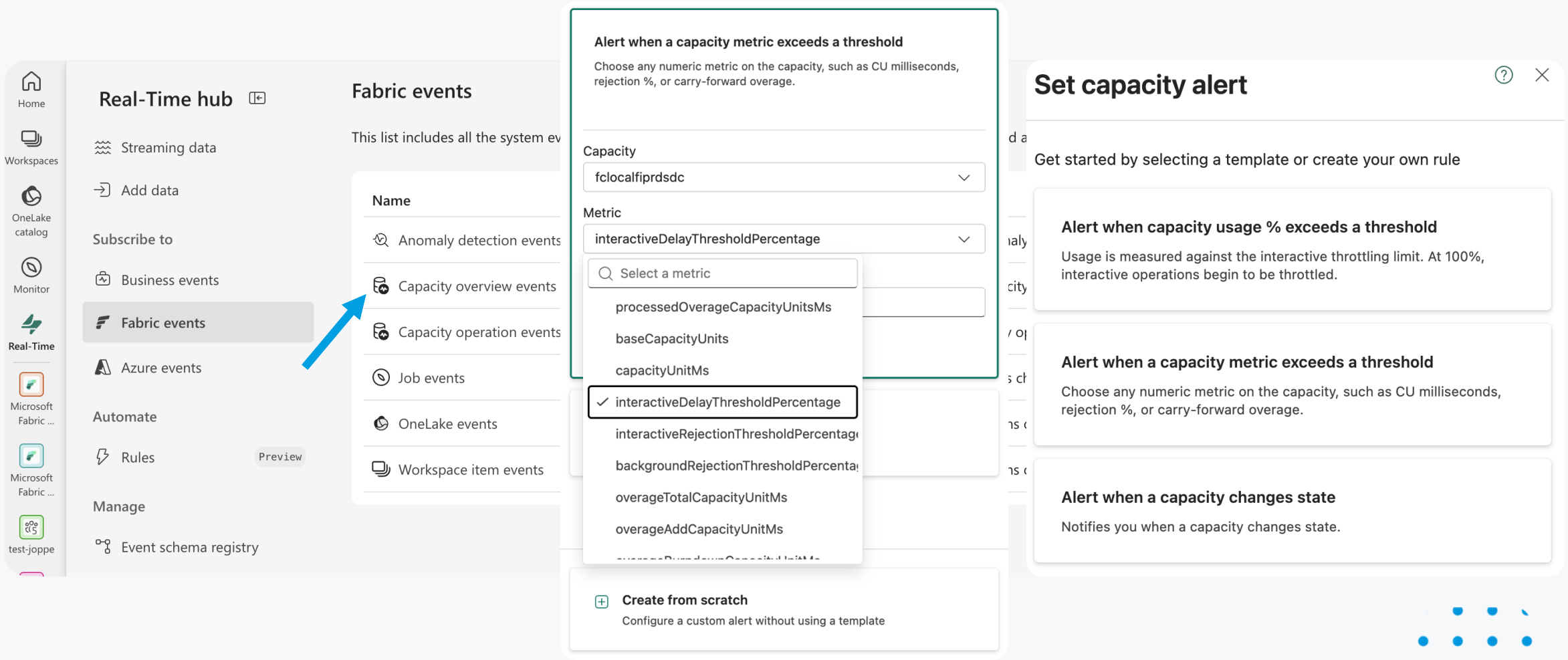
© Copyright DataSense

Monitoring Hub



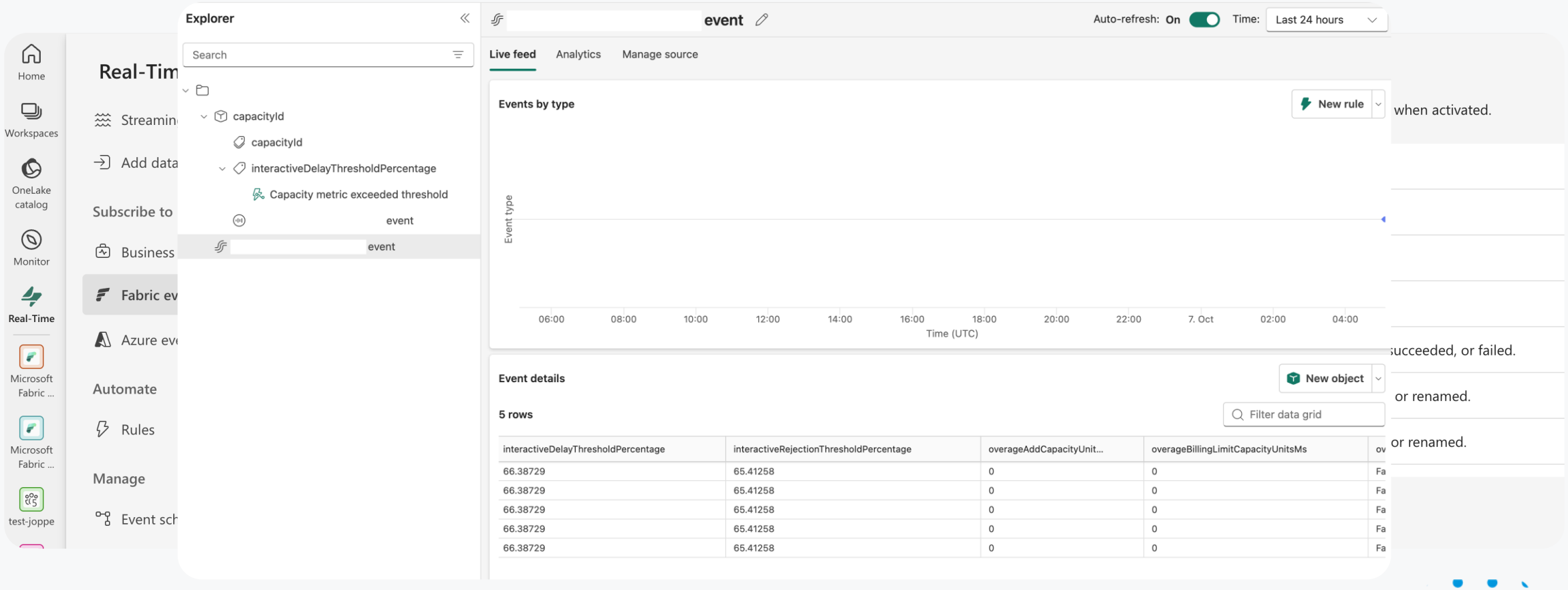
© Copyright DataSense

Real-Time hub



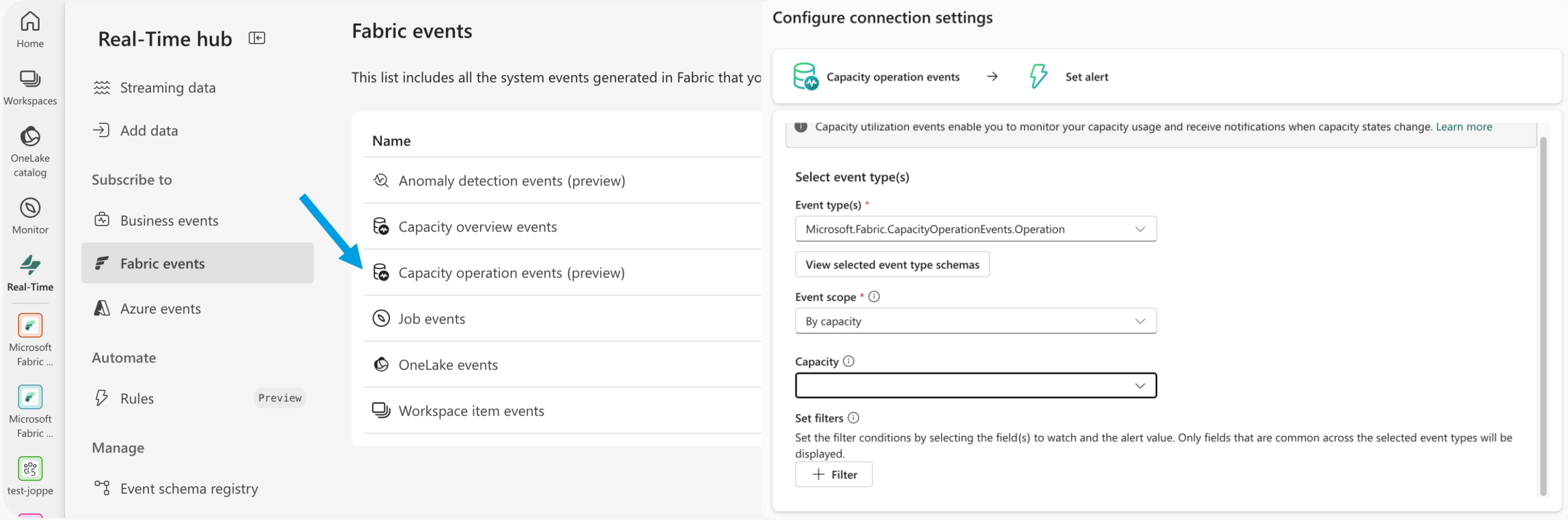
© Copyright DataSense

Real-Time hub



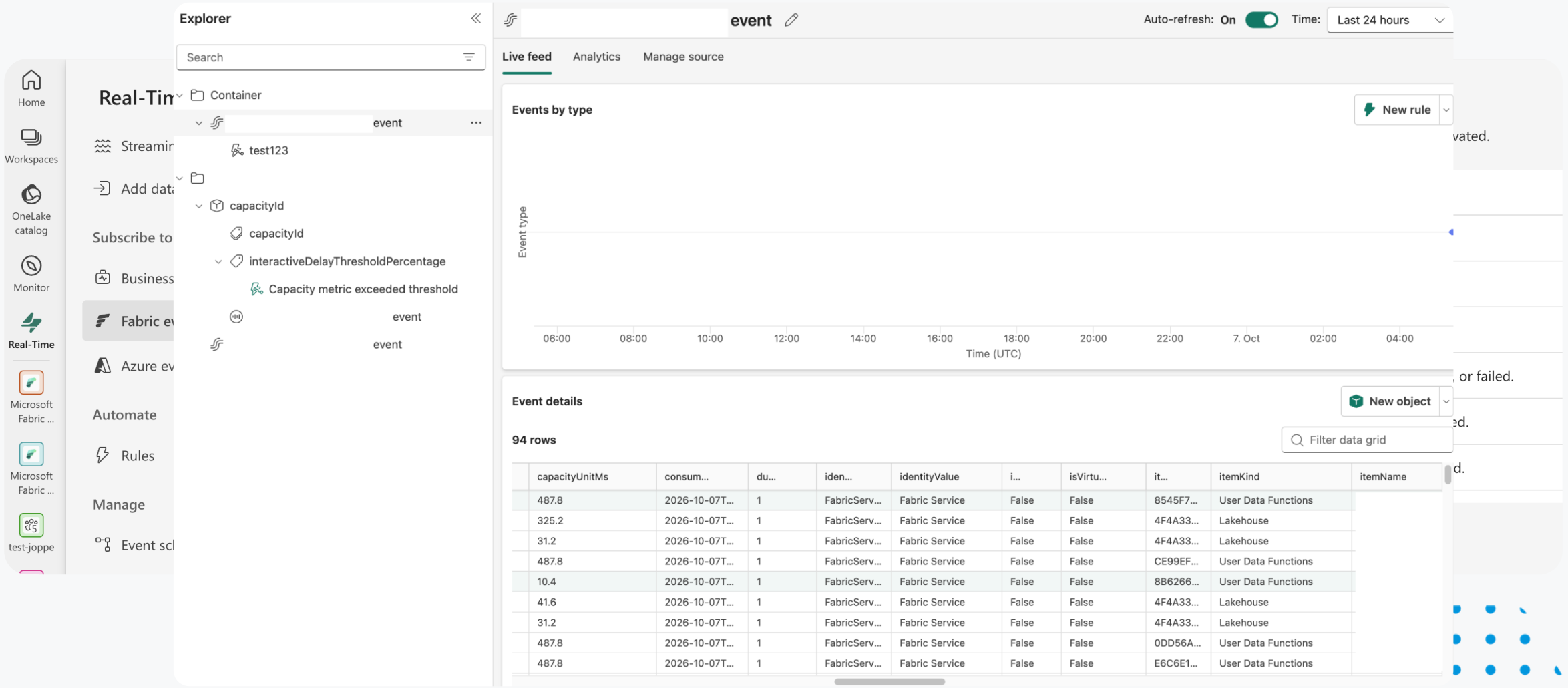
© Copyright DataSense

Real-Time hub



© Copyright DataSense

Real-Time hub



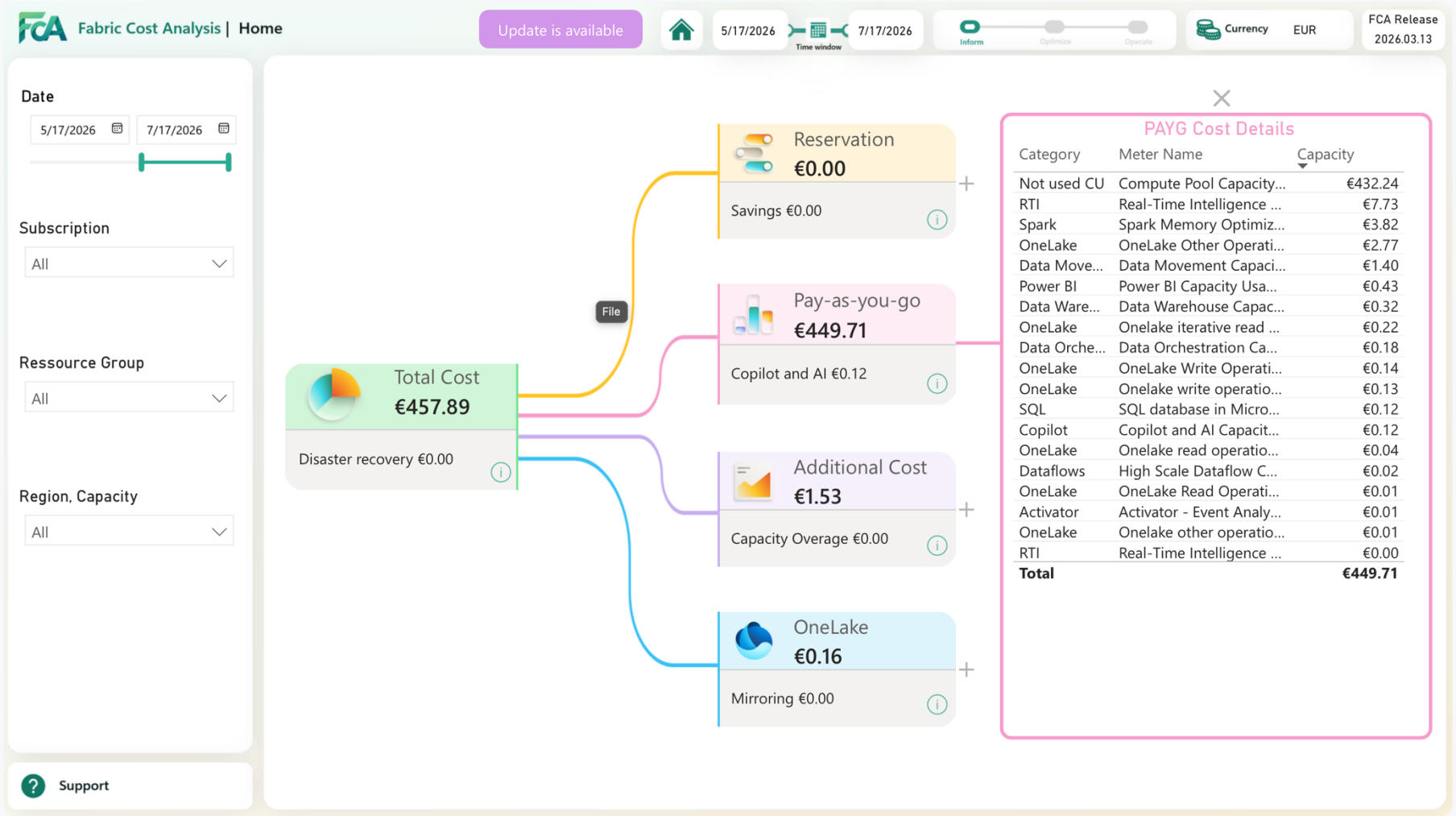
© Copyright DataSense

FUAM



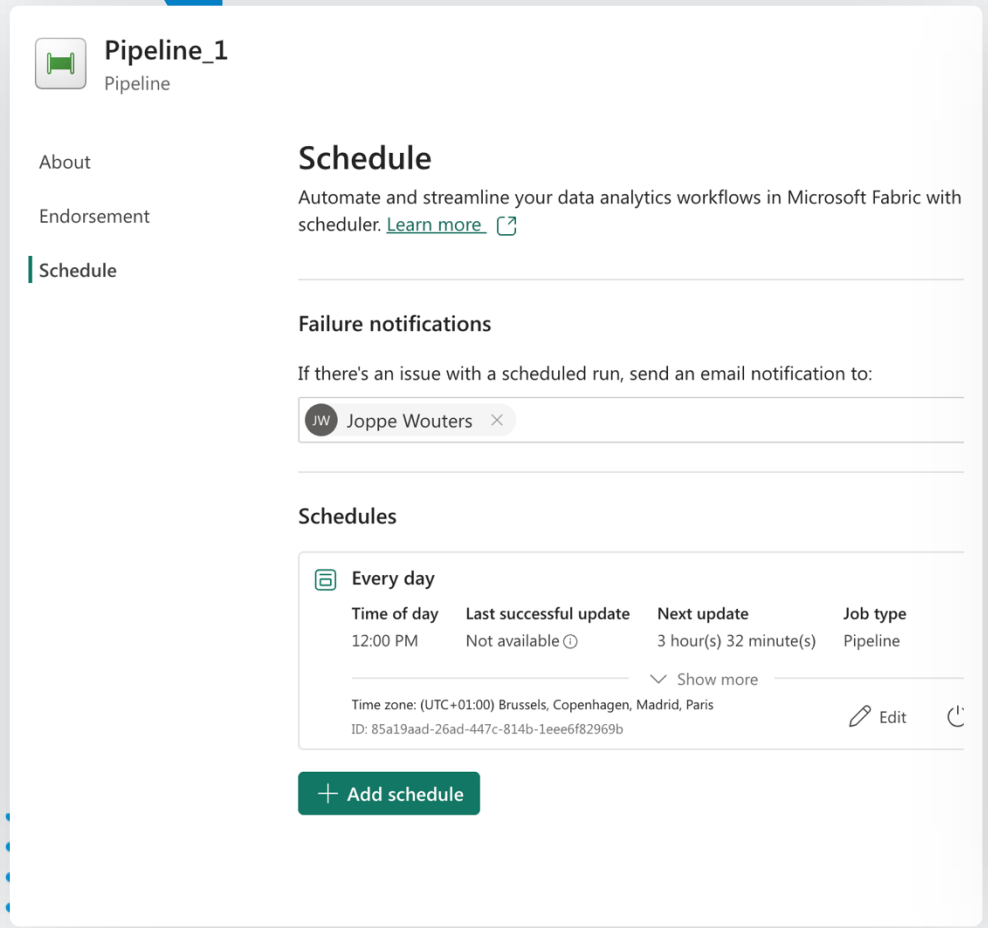
© Copyright DataSense

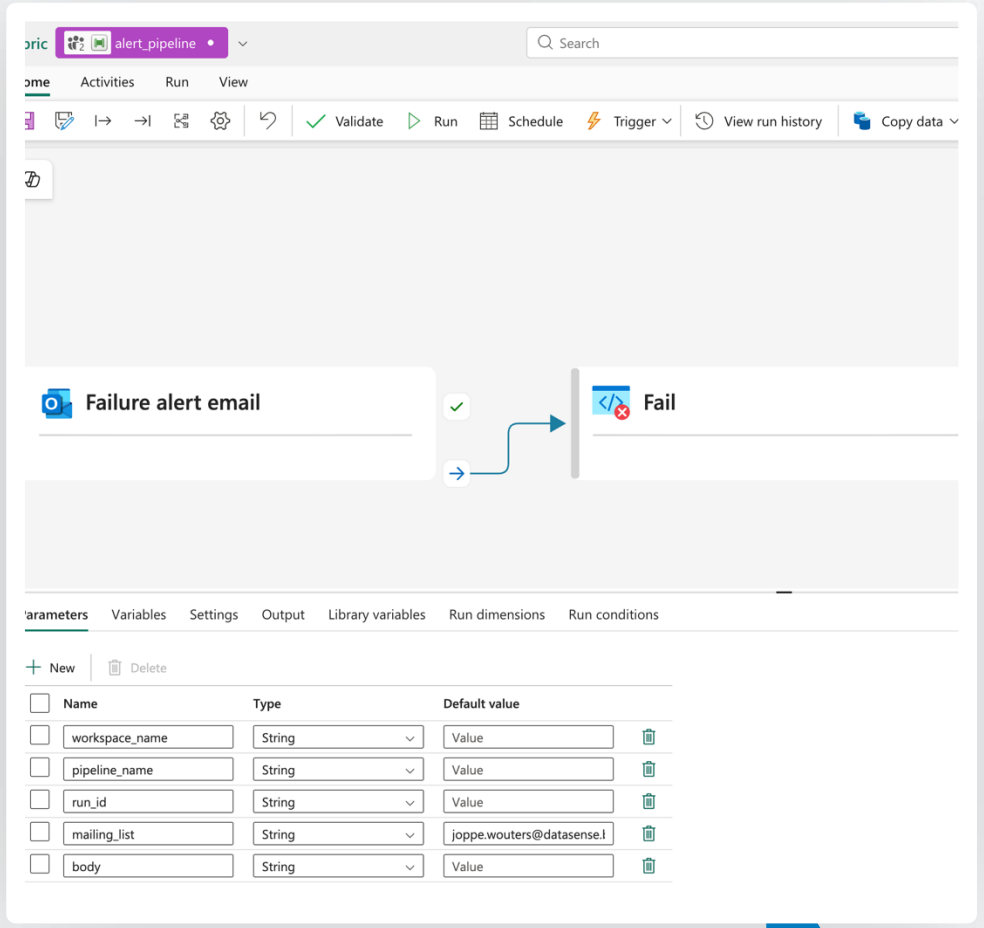
FCA



© Copyright DataSense

Pipeline failures



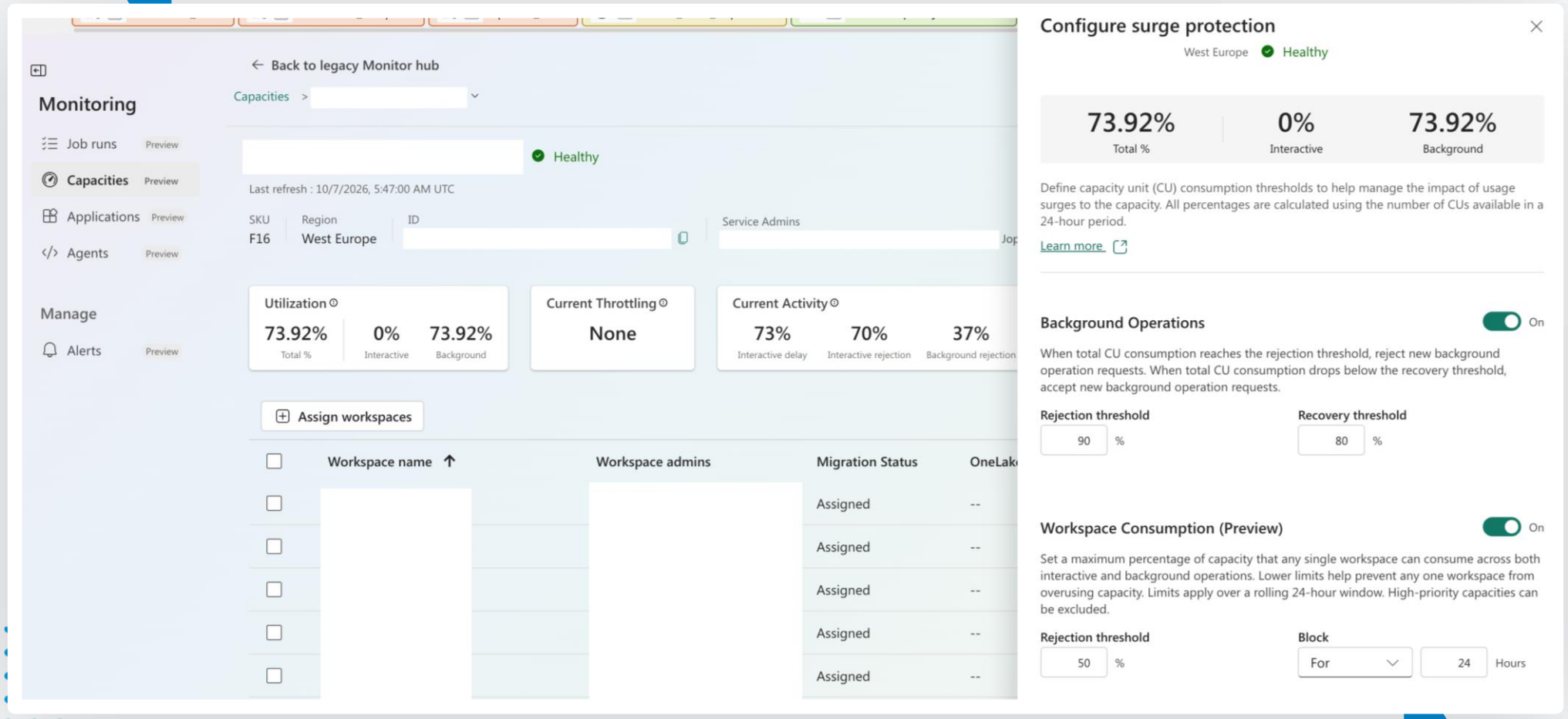


© Copyright DataSense



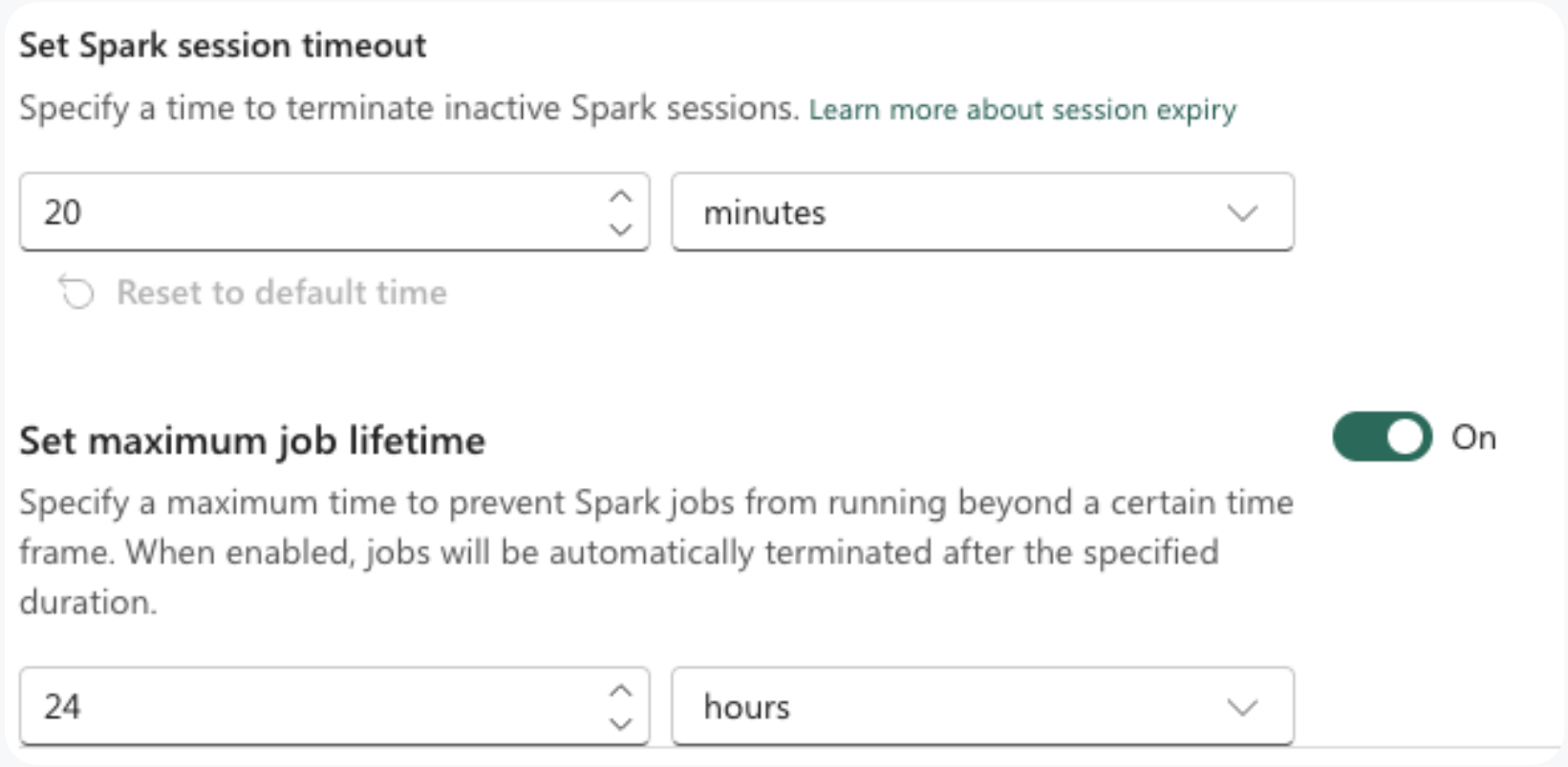
© Copyright DataSense

Surge protection



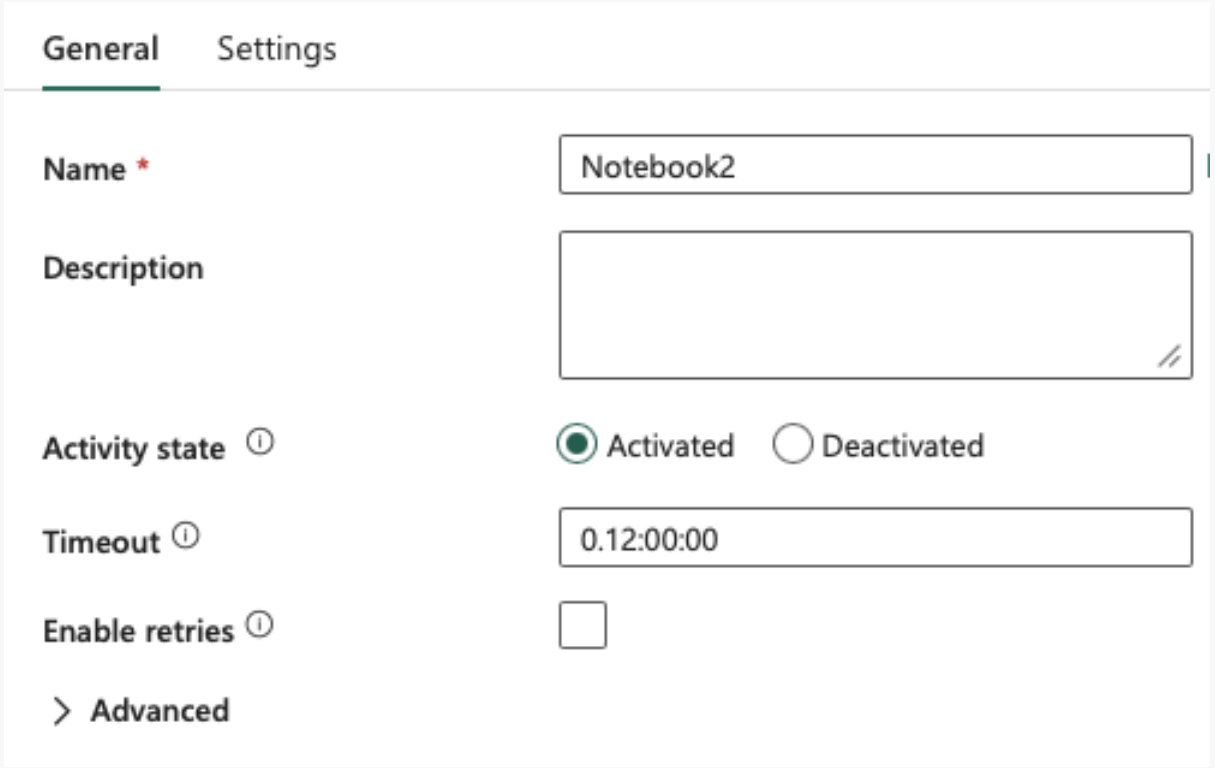
© Copyright DataSense

Workspace Spark settings



© Copyright DataSense

Pipeline activity timeouts



© Copyright DataSense

Policy set



New Policy set

An error occurred. Try again.

Was this error message helpful?

Name *

PolicySet_123

Location *

Policy set scope

A capacity

Fabric tenant

A capacity

© Copyright DataSense



© Copyright DataSense

Summary





© Copyright DataSense

Q & A



© Copyright DataSense

Stay in touch!

Joppe Wouters

www.linkedin.com/in/joppe-wouters

Joppe.wouters@datasense.be

www.datasense.be

© Copyright DataSense

Thank you, partners

















































© Copyright DataSense

Feedback



© Copyright DataSense